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*Evaluation of Feature Attribution Explanation models - effectiveness  
criteria and metrics for multi-objective models*

*Ocena skuteczności modeli opartych na atrybucji cech - kryteria  
skuteczności i metryki dla modeli wielokryterialnych*

|                   |                                    |
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# Abstract

Feature Attribution Explanation (FAE) methods are increasingly relied upon to make the decisions of deep neural networks intelligible, yet their evaluation remains fragmented: dozens of quality metrics exist, they frequently disagree, and it is rarely clear which of them should be trusted. This thesis treats the *evaluation pipeline*, rather than any new explanation method, as the unit of contribution and develops a multi-criteria framework for assessing FAE methods along six quality dimensions—Faithfulness, Robustness, Localization, Complexity, Randomization, and Axiomatic correctness—drawn from the Quantus taxonomy.

The work makes four contributions. First, it presents an empirical non-redundancy analysis of the available metrics, using correlation-based pruning to identify a compact, non-redundant representative subset. Second, it proposes a multi-criteria aggregation framework that integrates MetaQuantus-style meta-validation (Noise Resilience and Adversarial Reactivity) with an Autoweighted aggregation scheme inspired by NormEnsembleXAI, and introduces a novel MetaQuantus-discounted weighting operator that attenuates the influence of unreliable metrics. Third, it characterises empirically when aggregated rankings diverge from single-metric rankings and compares individual FAE methods against NormEnsembleXAI-ensembled attributions. Fourth, it validates the framework on the ISIC 2017 dermoscopic dataset, whose expert-annotated lesion masks enable Localization metrics, using Friedman, Nemenyi, and Wilcoxon tests to establish statistical significance.

Empirically, over two classifiers (ResNet-18 and SqueezeNet 1.1), seven attribution methods, and the 600-image ISIC 2017 test split, the framework yields four findings. A structural pre-screen and correlation-based redundancy pruning together reduce the twelve-metric battery to the *same* seven-metric non-redundant subset on both architectures, indicating that the redundancy structure is a property of the metric battery rather than of the classifier. The meta-evaluation identifies Max-Sensitivity, Sparseness, and Pixel-Flipping as the most reliable metrics and Faithfulness Correlation and Relevance Mass Accuracy as the least; the reliability-discounted weighting built on these scores reorders the ranking on one architecture and barely moves it on the other. The aggregated rankings prove weighting-sensitive in a model-dependent way: on ResNet-18 the mid-field reorders substantially between uniform and the dispersion-based Autoweighted weighting (Spearman  $\rho = 0.75$ ) while Occlusion stays first, whereas the SqueezeNet ranking is nearly stable; a single-metric ranking correlates only moderately with the multi-criteria index—a concrete demonstration that single-metric conclusions are unsafe. Finally, once ensembled and individual scores are normalised on a common scale, NormEnsembleXAI-ensembled

attributions beat none of the individual baselines: they are significantly worse than the average individual method on ResNet-18, statistically indistinguishable from it on SqueezeNet, and decisively below the best individual method on both models; the ranking differences are statistically significant throughout (Friedman and Wilcoxon tests,  $p < 10^{-3}$ ). Together these elements provide a reproducible, meta-validated procedure for comparing explanation methods under multiple, potentially conflicting effectiveness criteria.

**Keywords:** Explainable AI, feature attribution, evaluation metrics, meta-evaluation, multi-criteria aggregation, Quantus, MetaQuantus, NormEnsembleXAI, ISIC 2017, skin lesion classification.

## Streszczenie

Metody wyjaśniania oparte na atrybucji cech (ang. *Feature Attribution Explanations*, FAE) są coraz częściej wykorzystywane, aby uczynić decyzje głębokich sieci neuronowych zrozumiałymi, jednak ich ocena pozostaje rozproszona: istnieją dziesiątki metryk jakości, metryki te często wzajemnie się nie zgadzają i rzadko wiadomo, którym z nich należy ufać. Niniejsza praca traktuje jako przedmiot wkładu sam *potok ewaluacyjny*, a nie kolejną metodę wyjaśniania, i rozwija wielokryterialny framework oceny metod FAE wzdłuż sześciu wymiarów jakości—wierności (*Faithfulness*), odporności (*Robustness*), lokalizacji (*Localization*), złożoności (*Complexity*), randomizacji (*Randomization*) i poprawności aksjomatycznej—zaczepniętych z taksonomii Quantus.

Praca wnosi cztery wkłady. Po pierwsze, przedstawia empiryczną analizę nieredundancji dostępnych metryk, wykorzystując przycinanie oparte na korelacji do wyznaczenia zwartego, nieredundantnego podzbioru reprezentatywnego. Po drugie, proponuje wielokryterialny framework agregacji, który integruje meta-walidację w stylu MetaQuantus (odporność na szum oraz reaktywność adversarialną) ze schematem ważenia Autoweighted inspirowanym NormEnsembleXAI, oraz wprowadza nowy operator ważenia z dyskontem MetaQuantus, osłabiający wpływ niewiarygodnych metryk. Po trzecie, charakteryzuje empirycznie, kiedy rankingi zagregowane rozchodzą się z rankingami opartymi na pojedynczej metryce, i porównuje pojedyncze metody FAE z atrybucjami zespolonymi metodą NormEnsembleXAI. Po czwarte, waliduje framework na dermoskopowym zbiorze danych ISIC 2017, którego eksperckie maski zmian skórnych umożliwiają stosowanie metryk lokalizacji, wykorzystując testy Friedmana, Nemenyiego i Wilcozona do ustalenia istotności statystycznej.

Empirycznie, dla dwóch klasyfikatorów (ResNet-18 i SqueezeNet 1.1), siedmiu metod atrybucji oraz 600-obrazowego zbioru testowego ISIC 2017 framework przynosi cztery ustalenia. Wstępna selekcja strukturalna wraz z przycinaniem redundancji opartym na korelacji redukuje dwunastometrykową baterię do *tego samego* siedmioelementowego, nieredundantnego podzbioru na obu architekturach, co wskazuje, że struktura redundancji jest własnością baterii metryk, a nie klasyfikatora. Meta-ewaluacja wskazuje Max-Sensitivity, Sparseness i Pixel-Flipping jako metryki najbardziej wiarygodne, a Faithfulness Correlation i Relevance Mass Accuracy jako najmniej wiarygodne; oparte na tych wynikach ważenie z dyskontem wiarygodności przetasowuje ranking na jednej architekturze, a na drugiej niemal go nie zmienia. Zagregowane rankingi okazują się wrażliwe na schemat ważenia w sposób zależny od modelu: na ResNet-18 środek stawki istotnie się przestawia między ważeniem jednostajnym a opartym na rozrzucie ważeniem Autoweighted (współczynnik Spearmana  $\rho = 0,75$ ), podczas gdy Occlusion pozostaje na

pierwszym miejscu, a ranking dla SqueezeNet jest niemal stabilny; ranking oparty na pojedynczej metryce koreluje z indeksem wielokryterialnym jedynie umiarkowanie—co konkretnie pokazuje, że wnioski wyciągane z pojedynczej metryki są zawodne. Wreszcie, po znormalizowaniu wyników zespołu i metod indywidualnych na wspólnej skali atrybucje zespolone metodą NormEnsembleXAI nie pokonują żadnego z raportowanych punktów odniesienia: są istotnie gorsze od przeciętnej metody indywidualnej na ResNet-18, statystycznie nieodróżnialne od niej na SqueezeNet i wyraźnie gorsze od najlepszej metody indywidualnej na obu modelach; różnice rankingowe pozostają istotne statystycznie (testy Friedmana i Wilcoxona,  $p < 10^{-3}$ ). Łącznie elementy te dostarczają odtwarzalnej, meta-walidowanej procedury porównywania metod wyjaśniania w warunkach wielu, potencjalnie sprzecznych kryteriów skuteczności.

**Słowa kluczowe:** wyjaśnialna sztuczna inteligencja, atrybucja cech, metryki ewaluacji, meta-ewaluacja, agregacja wielokryterialna, Quantus, MetaQuantus, NormEnsembleXAI, ISIC 2017, klasyfikacja zmian skórnych.

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# 1. Introduction

The evolution of Deep Learning (DL) has produced a paradigm shift in artificial intelligence in which predictive performance frequently comes at the cost of transparency, a tension commonly described as the interpretability–performance trade-off [1]. Modern neural networks routinely match or exceed human accuracy on perceptual tasks, yet their internal decision processes remain opaque “black boxes” whose outputs cannot be inspected directly [2]. The field of Explainable AI (XAI) responds to this opacity with a family of Feature Attribution Explanation (FAE) methods that assign an importance score to each input feature, producing saliency or attribution maps intended to reveal which parts of an input drove a given prediction [3, 4]. As these methods migrate from research prototypes into high-stakes deployments, the question of whether a particular explanation can actually be trusted has become as important as the question of whether a model is accurate. This thesis addresses that question directly: rather than proposing a new attribution method, it treats the *evaluation* of attribution methods as the central problem and develops a multi-criteria framework for assessing FAE methods rigorously and reproducibly.

## 1.1. Research motivation

The motivation for this work is driven by three interrelated factors: a legal and ethical imperative for auditable AI, a methodological crisis in how explanations are evaluated, and the particular difficulty of establishing ground truth for explanations in the medical domain.

### 1.1.1. The legal and ethical imperative

Transparency in automated decision-making is no longer merely a desirable technical property; it has become a regulatory requirement. The European Union’s Artificial Intelligence Act mandates that high-risk AI systems be designed and developed so that their operation is sufficiently transparent to enable deployers to interpret a system’s output and use it appropriately [5]. The General Data Protection Regulation reinforces this trajectory by establishing provisions, frequently summarised as a “right to explanation”, that entitle individuals to meaningful information about the logic of automated decisions that significantly affect them [6]. Medical diagnostics is a paradigmatic high-risk setting: a classifier that distinguishes malignant from benign skin lesions cannot be adopted clinically unless its reasoning can be examined and audited, because a physician cannot responsibly defer to a prediction whose basis cannot

be verified [7]. Crucially, the regulatory demand for transparency is only meaningful if the explanations supplied to satisfy it are themselves trustworthy. An attribution map that is plausible-looking but unfaithful to the model offers a false sense of accountability and may be worse than no explanation at all. Work on operationalising the EU AI Act confirms that translating its obligations into concrete XAI design and evaluation choices remains an open problem, precisely in clinical decision-support settings [8]. This observation shifts the burden from *producing* explanations to *evaluating* them, and motivates the need for quantitative, defensible criteria by which an explanation’s quality can be judged.

### 1.1.2. The crisis of evaluation

Despite a rapid proliferation of XAI methods, the field suffers from what has been termed a crisis of evaluation: there is little consensus on how to measure whether an explanation is good, and the metrics that do exist often disagree [9]. Foundational work has shown that several widely used attribution methods fail elementary sanity checks, producing visually convincing maps that remain essentially unchanged even when the parameters of the model being explained are randomised, which means those maps cannot be reflecting the learned function at all [2]. Compounding this, different methods frequently attribute importance to different features for the same prediction, a phenomenon documented as systematic disagreement across explanation methods [10]; no single estimator family dominates across the quality criteria—gradient-based estimators, for example, are efficient and often faithful yet notoriously unstable under small input perturbations, while perturbation-based estimators carry their own sampling-induced instability and cost [11]. A natural response to such single-estimator unreliability is to combine several attribution methods into an ensemble whose aggregated map is more stable than any of its constituents, as in the NormEnsembleXAI approach [11].

Ensembling, however, only relocates the evaluation problem rather than solving it, because one must still decide *which* quality criteria to optimise and *how* to weigh them. A large and heterogeneous catalogue of evaluation metrics now exists, organised by toolkits such as Quantus into categories spanning Faithfulness, Robustness, Localization, Complexity, Randomization, and Axiomatic correctness [9, 12]. These categories are not independent, and some metrics within a category are highly correlated, so naively reporting many scores can give an illusion of thoroughness while measuring the same underlying property repeatedly [13]. Worse, the metrics themselves are not equally reliable: a metric can be insensitive to perturbations it ought to detect, or reactive to changes it ought to ignore, and meta-evaluation procedures have been developed precisely to quantify this reliability [14, 15]. Evaluating FAE methods responsibly therefore requires a framework that (i) prunes redundant metrics, (ii) discounts metrics that are themselves unreliable, and (iii) aggregates the remaining, validated criteria into a single comparable effectiveness signal. Because this aggregation balances several conflicting quality dimensions simultaneously, the evaluation is inherently *multi-objective* in the space of metrics, even though the predictive models being explained are ordinary single-output classifiers.

### 1.1.3. Ground truth in the medical domain

A further difficulty specific to high-stakes vision tasks is that most metrics of the Localization family presuppose a spatial reference against which an attribution map can be compared, yet such ground truth is rarely available. General-purpose image benchmarks supply only image-level class labels, leaving no principled way to ask whether an explanation highlights the *semantically correct* region. The medical-imaging setting alleviates this problem in a way that aligns naturally with the regulatory motivation above: expert-annotated lesion masks delineate exactly the region a clinically valid explanation ought to emphasise, and these masks therefore serve as an objective spatial reference for Localization metrics without any additional labelling effort [7]. The ISIC 2017 dermoscopic dataset [16], used throughout this thesis, provides such expert masks paired with each image, which makes it possible to ground the Localization category empirically and gives the evaluation framework a concrete, auditable, and ethically motivated testbed.

## 1.2. Research contributions

The unit of novelty in this thesis is the evaluation pipeline—the procedure by which feature attribution methods are scored, validated, and ranked—rather than any new explanation method or predictive architecture. Against the background outlined above, the contributions of this work are as follows.

1. **Non-redundancy analysis of FAE evaluation metrics.** An empirical correlation study across the six Quantus metric categories that identifies highly correlated metrics within each category and derives a compact, non-redundant representative subset, so that subsequent evaluation measures distinct quality dimensions rather than re-measuring the same property [9, 13].
2. **A multi-criteria evaluation framework, and a test of reliability-discounted aggregation.** A methodological framework that integrates MetaQuantus-style meta-validation—quantifying each metric’s Noise Resilience and Adversarial Reactivity—with an Autoweighted aggregation scheme inspired by NormEnsembleXAI, and that formulates and empirically tests a MetaQuantus-discounted extension of Autoweighted aggregation in which the influence of a metric is attenuated in proportion to its measured unreliability, in order to determine *whether* such a reliability discount changes the resulting FAE ranking [14, 11].
3. **Empirical characterization of metric ensembling.** An analysis of when aggregation of metric scores produces FAE rankings that diverge from single-metric rankings, when it does not, and which metric families benefit most from aggregation, thereby clarifying the practical value added by multi-criteria evaluation over reporting individual metrics in isolation [9].

4. **Empirical validation on ISIC 2017 with statistical testing.** A practical assessment of the framework on the ISIC 2017 skin-lesion classification task, whose expert-annotated masks enable Localization metrics, comparing individual FAE methods—and individual versus NormEnsembleXAI-ensembled attributions—under the proposed aggregation, with statistical significance established through Friedman and Nemenyi tests across methods and Wilcoxon tests for paired comparisons [16, 11].

### 1.3. Thesis contents

The remainder of this thesis is structured as follows.

- **Chapter 2: Literature Review** provides an overview of the state of the art in Feature Attribution Explanation methods, detailing the taxonomy of gradient-based [3] and perturbation-based [4] approaches. It establishes the crisis of evaluation [9] by discussing the failure of common saliency maps to pass model-sensitivity sanity checks [2], and it defines the mathematical foundations of the atomic metrics—Faithfulness, Robustness, Localization, Complexity, and related families—that serve as the building blocks for modular evaluation [9, 17].
- **Chapter 3: Methodology** introduces the proposed multi-criteria evaluation framework. It describes how heterogeneous attribution outputs are aligned through Second Moment Scaling normalization and combined, where ensembling is applied, via the NormEnsembleXAI aggregation functions [11]; it specifies the metric set across the six Quantus categories and the redundancy analysis used to prune it; and it details the Autoweighted aggregation together with the MetaQuantus-discounted weighting that down-weights unreliable metrics, whose effect on the ranking is then tested empirically [14, 11].
- **Chapter 4: Experiments and Results** presents the empirical validation of the framework on the ISIC 2017 dermoscopic dataset [16], whose expert-annotated lesion masks enable Localization metrics. It reports the metric redundancy analysis, the MetaQuantus-based reliability of each metric, the comparison of uniform, Autoweighted, and MetaQuantus-discounted aggregation, the contrast between individual and NormEnsembleXAI-ensembled attributions [11], and the statistical significance of the resulting rankings via Friedman, Nemenyi, and Wilcoxon testing.
- **Chapter 5: Discussion** interprets the rankings, examines the threats to validity—notably the sample-size limitation of the redundancy and reliability estimates—and delimits the scope within which the conclusions generalise.
- **Chapter 6: Conclusions and Future Work** summarises the findings against each contribution, discusses the trade-offs between interpretability and performance [1] and the computational cost of multi-criteria evaluation [11], and outlines future research into automated metric validation and explanation auditing within the scope of the EU AI Act.

## 2. Literature Review

The field of Explainable Artificial Intelligence (XAI) has expanded rapidly in response to the opacity of modern Deep Learning (DL) architectures. While early research focused primarily on generating visually appealing explanations, recent scholarship has shifted towards rigorous quantification of explanation quality. This chapter provides a taxonomy of Feature Attribution Explanation (FAE) methods, analyzes the "Crisis of Evaluation" precipitated by the lack of ground truth, and systematizes the mathematical frameworks used to measure Faithfulness, Robustness, and Complexity. Finally, it reviews the theoretical foundations of XAI Ensembling as a mechanism to mitigate the limitations of individual estimators.

### 2.1. Feature Attribution Explanation (FAE) Methods

Feature attribution methods, often referred to as saliency maps in computer vision, aim to assign a scalar relevance score  $R_i$  to each input feature  $x_i$  (e.g., a pixel), representing its contribution to the model's output  $f(x)$  for a specific target class  $c$ . Formally, an explanation method  $\Phi$  takes a model  $f$  and an input  $x$  to produce a map  $E = \Phi(f, x)$ , where  $E \in \mathbb{R}^d$  and  $d$  is the dimensionality of the input.

Current literature broadly categorizes these methods based on their access to the model's internal parameters: Gradient-based (White-box) and Perturbation-based (Black-box) approaches [9].

#### 2.1.1. Gradient-based Methods (White-box)

Gradient-based methods exploit the differentiable nature of neural networks. They require access to the model's weights and gradients, operating on the intuition that the gradient of the output with respect to the input represents the model's sensitivity to feature changes. The methods below form a progression: from raw gradients, to path-integrated gradients that restore axiomatic guarantees, to feature-space gradients that trade pixel resolution for class-discriminative localization.

##### 2.1.1.1. Saliency Maps

The foundational approach, proposed by Simonyan et al., computes the gradient of the class score  $S_c$  with respect to the input image  $I$  [18]. The relevance map is defined as:

$$E_{Vanilla} = \left| \frac{\partial S_c}{\partial I} \right| \quad (2.1)$$

While computationally efficient, vanilla gradients often suffer from "shattered gradients," resulting in noisy visualizations that lack class-discriminateness. Furthermore, they are susceptible to the "saturation problem," where a feature may strongly contribute to a prediction, but its gradient is zero because the activation is saturated (e.g., in a ReLU function) [19].

### 2.1.1.2. Integrated Gradients (IG)

To address gradient saturation, Sundararajan et al. introduced Integrated Gradients, which fundamentally relies on the *Axiom of Completeness*. IG accumulates gradients along a linear path from a baseline input  $x'$  (usually a black image) to the actual input  $x$  [19]. The attribution for the  $i$ -th feature is given by:

$$IG_i(x) = (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{\partial f(x' + \alpha(x - x'))}{\partial x_i} d\alpha \quad (2.2)$$

This method ensures that the sum of attributions equals the difference between the model output at the input and the baseline, providing a theoretical guarantee missing in vanilla gradients.

### 2.1.1.3. Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) moves away from pixel-space gradients to feature-space gradients. It utilizes the gradients flowing into the final convolutional layer of a CNN to produce a coarse localization map [3]. Global Average Pooling is applied to the gradients to obtain neuron importance weights  $\alpha_k^c$ :

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (2.3)$$

where  $A^k$  is the  $k$ -th feature map. The final map is a ReLU-activated linear combination of these maps. Grad-CAM is highly effective for localization but lacks fine-grained pixel detail, often necessitating combination with Guided Backpropagation (Guided Grad-CAM).

## 2.1.2. Perturbation-based Methods (Black-box)

Perturbation-based methods operate under a model-agnostic paradigm. They estimate feature importance by systematically modifying (perturbing) the input and observing the resulting changes in the model's output prediction. Because they treat the model as a black box, they apply uniformly across architectures, but their reliance on sampling makes them both computationally expensive and, as noted below, susceptible to estimation noise. The thesis surveys LIME and SHAP here for completeness, but—as Chapter 3 details—the final evaluation panel retains only the gradient-based methods together with Occlusion, since the Robustness metrics re-invoke the explanation function for every perturbation sample, multiplying the already-substantial surrogate-fitting cost of LIME and KernelSHAP beyond what the scale of the full study permits.

### 2.1.2.1. LIME (Local Interpretable Model-agnostic Explanations)

LIME creates a surrogate model  $g$  (e.g., a linear regression) that is locally faithful to the complex model  $f$  around the instance  $x$  [4]. It generates a neighborhood of perturbed samples  $\mathcal{Z}$  weighted by a

proximity kernel  $\pi_x$ . The explanation is found by minimizing the loss  $\mathcal{L}$ :

$$\xi(x) = \operatorname{argmin}_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \quad (2.4)$$

where  $\Omega(g)$  penalizes the complexity of the explanation. While LIME is versatile, its reliance on random sampling introduces instability, where repeated runs on the same input can yield different explanations [11].

### 2.1.2.2. SHAP (SHapley Additive exPlanations)

SHAP adapts cooperative game theory to XAI, treating features as players in a coalition. The attribution of a feature is its Shapley value - the average marginal contribution of the feature value across all possible feature coalitions [20]. The SHAP value  $\phi_i$  is defined as:

$$\phi_i(f, x) = \sum_{z' \subseteq x' \setminus \{i\}} \frac{|z'|! (M - |z'| - 1)!}{M!} [f_x(z' \cup \{i\}) - f_x(z')] \quad (2.5)$$

SHAP values are the unique additive attribution satisfying *Local Accuracy*, *Missingness*, and *Consistency* [20], thereby inheriting the classical Shapley axioms of Efficiency, Symmetry, Dummy, and Additivity. However, exact computation is exponential in the number of features, leading to approximation methods like GradientSHAP or KernelSHAP, which reintroduce estimation noise.

### 2.1.3. Localization Metrics

Localization metrics assess whether the regions highlighted by an attribution map correspond to semantically meaningful areas of the input. Unlike Faithfulness metrics, which probe the model's response to perturbation, Localization metrics require an external ground-truth mask delineating the object or region of interest. Two widely adopted metrics in this family are the Pointing Game and Relevance Mass Accuracy.

#### 2.1.3.1. Pointing Game

The Pointing Game evaluates whether the single most salient pixel in the attribution map falls within the annotated region of interest. Given an explanation  $E$  and a binary segmentation mask  $\mathcal{M}$ , the Pointing Game score for a single image is defined as:

$$\text{PG} = \begin{cases} 1 & \text{if } \operatorname{argmax}_i |E_i| \in \mathcal{M}, \\ 0 & \text{otherwise.} \end{cases} \quad (2.6)$$

The metric is typically averaged across the test set to produce a hit rate. Its binary nature makes it interpretable but coarse: a method that assigns high attribution to many pixels inside the mask but whose global maximum lies just outside receives a score of zero. The Pointing Game in its present form was introduced by Zhang et al. [21] as part of the Excitation Backprop study, building on the class-activation-mapping line of work that first established the use of localization maps for weakly supervised tasks.

### 2.1.3.2. Relevance Mass Accuracy

Relevance Mass Accuracy (RMA) captures the fraction of total positive attribution mass that falls within the segmentation mask:

$$\text{RMA} = \frac{\sum_{i \in \mathcal{M}} \max(E_i, 0)}{\sum_i \max(E_i, 0)}. \quad (2.7)$$

RMA ranges from 0 (no positive attribution inside the mask) to 1 (all positive attribution is localized). Unlike the Pointing Game, RMA evaluates the entire attribution distribution rather than a single pixel, providing a smoother and more discriminative signal [9].

### 2.1.3.3. Role of Segmentation Masks

Both Pointing Game and RMA require a binary segmentation mask aligned with the input image. This requirement limits Localization evaluation to datasets that provide pixel-level annotations. The ISIC 2017 dataset (Task 1) includes expert-annotated lesion boundaries for every image, enabling Localization assessment without any additional labeling effort. Many commonly used benchmarks lack such masks, which is one reason Localization metrics are underreported in the literature despite their intuitive appeal.

## 2.2. The Crisis of Evaluation

A critical challenge in XAI research is the absence of "ground truth." Unlike classification tasks where a label  $y$  is known, there is rarely a verified map of which pixels a neural network *actually* used. This ambiguity led to a reliance on "Plausibility" - visual assessment by humans - which proved deceptive.

A seminal study by Adebayo et al., titled "Sanity Checks for Saliency Maps," demonstrated that visual quality does not imply faithfulness [2]. They introduced the **Model Parameter Randomization Test**, where model weights are progressively randomized from the output layer to the input layer. A valid explanation method should degrade to noise as the model's knowledge is destroyed.

- **Failure of Guided Backpropagation:** The study revealed that Guided Backpropagation and Guided Grad-CAM produced nearly identical heatmaps for a trained model and a random network. This indicates these methods act largely as edge detectors, independent of the model's learned representations.
- **Implication:** This finding established a "Crisis of Evaluation" [9], necessitating a shift from human-grounded evaluation (which measures trust/plausibility) to functionally-grounded evaluation (which measures mathematical properties like fidelity).

## 2.3. Taxonomy of Evaluation Metrics

To rigorously assess FAE models, recent literature proposes a modular taxonomy of metrics. This thesis adopts the framework formalized by Quantus [9], covering all six of its categories: Faithfulness, Robustness, Localization (Section 2.1.3), Complexity, Randomization, and Axiomatic correctness.

### 2.3.1. Faithfulness (Fidelity)

Faithfulness quantifies how accurately the explanation reflects the model’s internal decision logic, regardless of human intelligibility.

#### 2.3.1.1. Pixel Flipping (AUC)

Pixel Flipping, introduced by Bach et al. [22] and generalized to region perturbation with an area-under-the-curve protocol by Samek et al. [23], evaluates faithfulness by iteratively removing the most important features (as ranked by the attribution map) and measuring the resulting degradation in the model’s output. Pixels are replaced in descending order of attribution magnitude with a baseline value (e.g., black), producing a curve of model outputs as ever more of the highest-ranked features are removed. The metric summarizes this degradation curve by its area under the curve:

$$\text{PixelFlipping}(x) = \frac{1}{N} \sum_{k=1}^N f(x^{(k)}), \quad (2.8)$$

where  $x^{(k)}$  is the image with the top  $k$  most important features replaced. A faithful attribution identifies the truly critical features first, so the prediction collapses early and the area under the degradation curve is small: *lower* scores are better. The Quantus implementation groups features into steps of configurable size (`features_in_step`), making the metric tractable for high-resolution inputs [9].

This formulation generalizes Deletion and Insertion curves [24]: Deletion corresponds to removal in descending attribution order, while Insertion reverses the process by adding features to a blank canvas.

A structural caveat applies to this whole family of perturbation-based metrics: replacing pixels with a baseline produces inputs that lie off the training distribution, so part of the measured prediction drop may reflect the model’s reaction to out-of-distribution inputs rather than the quality of the attribution. Recent work addresses this confound explicitly—the F-Fidelity framework removes it via fine-tuning and random masking, recovering ground-truth explainer rankings that the naive protocol distorts [25].

#### 2.3.1.2. Faithfulness Correlation

Faithfulness Correlation, as implemented in Quantus, measures the Pearson correlation between attribution magnitude and the change in model output when features are masked in random subsets [9]. Given `nr_runs` randomly sampled subsets  $S_1, \dots, S_R$  of features:

$$\text{FC} = r \left( \left\{ \sum_{i \in S_j} E_i \right\}_{j=1}^R, \left\{ f(x) - f(x_{\setminus S_j}) \right\}_{j=1}^R \right), \quad (2.9)$$

where  $r(\cdot, \cdot)$  denotes the Pearson correlation coefficient and  $x_{\setminus S_j}$  is the input with features in  $S_j$  replaced by a baseline. A high positive correlation ( $FC \rightarrow +1$ ) indicates that regions with large attribution values are indeed the regions whose removal most degrades the model’s prediction.

A related measure, Infidelity [26], computes the expected squared error  $\mathbb{E}[(I^T E - \Delta f)^2]$  between the attribution-weighted perturbation and the actual output change. Although theoretically motivated, Infidelity is not used in the evaluation framework of this thesis; Faithfulness Correlation provides a comparable signal while being more interpretable and directly comparable across methods.

### 2.3.2. Robustness (Stability)

Robustness ensures that slight, non-semantic changes to the input (e.g., Gaussian noise) do not cause disproportionate changes in the explanation. This is crucial for user trust and defense against adversarial attacks.

The theoretical foundation for explanation robustness is Lipschitz continuity: a method  $\Phi$  is robust if the ratio  $\|\Phi(x) - \Phi(x')\| / \|x - x'\|$  remains bounded for inputs  $x'$  in a neighborhood of  $x$  [17]. The metrics adopted in this thesis operationalize this principle via sensitivity analysis under random perturbations.

#### 2.3.2.1. Max-Sensitivity

Max-Sensitivity, proposed by Yeh et al. [26] in “On the (In)fidelity and Sensitivity of Explanations,” measures the worst-case explanation change under small random perturbations of the input,  $\text{SENS}_{\text{MAX}} = \max_{\|\delta\| \leq r} \|\Phi(x + \delta) - \Phi(x)\|$ . The Quantus implementation used in this thesis estimates it from  $N$  perturbation samples  $\delta_1, \dots, \delta_N$  drawn uniformly with magnitude bounded by `lower_bound`, and reports the change relative to the unperturbed attribution:

$$\text{MaxSens}(\Phi, x) = \max_{j=1, \dots, N} \frac{\|\Phi(x + \delta_j) - \Phi(x)\|_F}{\|\Phi(x)\|_F}. \quad (2.10)$$

The implementation uses `nr_samples` = 10 perturbation draws with `lower_bound` = 0.2 controlling the perturbation magnitude. Lower values indicate a more stable explanation method.

#### 2.3.2.2. Avg-Sensitivity

Avg-Sensitivity replaces the maximum with the mean across the same set of perturbations:

$$\text{AvgSens}(\Phi, x) = \frac{1}{N} \sum_{j=1}^N \frac{\|\Phi(x + \delta_j) - \Phi(x)\|_F}{\|\Phi(x)\|_F}. \quad (2.11)$$

Lower is better. Together, Max-Sensitivity and Avg-Sensitivity provide complementary views: the former captures the tail behavior (susceptibility to worst-case perturbations), while the latter captures the expected behavior. Both are implemented in Quantus [9] and share hyperparameters in this thesis for comparability.

### 2.3.3. Complexity and Sparsity

Complexity metrics adhere to the principle of Occam’s Razor: simpler explanations are generally preferred for human cognition.

- **Sparseness:** Quantifies the concentration of the attribution distribution using a Gini-like coefficient. A perfectly sparse explanation assigns all relevance to a single feature (score  $\rightarrow 1$ ), while a uniform explanation scores 0. Higher is better.
- **Complexity:** Calculated as the Shannon entropy of the normalized absolute attribution distribution. High entropy indicates a diffuse explanation, while low entropy indicates a focused, sparse explanation. Lower is better.

Both metrics are computed directly from the attribution map without model access, making them applicable to any FAE method regardless of its generation mechanism [9].

### 2.3.4. Randomization

Randomization tests assess whether an explanation method genuinely relies on learned model parameters rather than spurious input statistics. If an explanation remains unchanged after the model’s parameters are destroyed, the method is not faithfully reflecting the model’s decision process.

#### 2.3.4.1. Model Parameter Randomisation Test (MPRT)

Proposed by Adebayo et al. as part of the “Sanity Checks for Saliency Maps” study [2], MPRT progressively randomizes the model’s weights layer by layer, proceeding from the output layer toward the input (top-down order). After each randomization step, the explanation is recomputed and its similarity to the original explanation is measured via Spearman or Pearson correlation. The metric reports the average correlation across all randomization depths:

$$\text{MPRT}(\Phi) = \frac{1}{L} \sum_{\ell=1}^L \rho(\Phi(f, x), \Phi(f^{(\ell)}, x)), \quad (2.12)$$

where  $f^{(\ell)}$  denotes the model with the top  $\ell$  layers randomized and  $\rho$  is the chosen correlation measure. Lower average correlation indicates a better method—one that is genuinely sensitive to the model’s learned parameters. As discussed in Section 2.2, Guided Backpropagation and Guided Grad-CAM fail this test, producing nearly identical maps for trained and randomized models.

#### 2.3.4.2. Random Logit Test

The Random Logit test complements MPRT by perturbing the explanation target rather than the model weights. The target class index is replaced with a randomly sampled alternative class, and the resulting explanation is compared to the original. The metric computes the correlation between the two explanation maps:

$$\text{RandomLogit}(\Phi) = \rho(\Phi(f, x, c), \Phi(f, x, c')), \quad c' \neq c. \quad (2.13)$$

A lower correlation indicates that the explanation is genuinely class-discriminative—it changes when the target class changes—rather than merely reflecting generic input features. Both MPRT and Random Logit are implemented in Quantus [9].

### 2.3.5. Axiomatic

Axiomatic metrics verify whether explanations satisfy formal mathematical axioms that a “correct” attribution should obey. These axioms provide theoretical guarantees that hold independently of the specific model or input.

#### 2.3.5.1. Completeness

The Completeness axiom, formalized by Sundararajan et al. [19], requires that the sum of individual feature attributions equals the difference between the model output at the input  $x$  and a baseline  $x'$ :

$$\left| \sum_i E_i - (f(x) - f(x')) \right| \approx 0. \quad (2.14)$$

The metric measures the absolute deviation from this axiom; lower is better. Completeness is only meaningful for methods that explicitly satisfy or approximate the completeness property by construction—specifically Integrated Gradients, DeepLift, and LRP. For other methods, the metric returns NaN and is excluded from aggregation.

#### 2.3.5.2. Non-Sensitivity

Non-Sensitivity measures how much attribution a method assigns to features that are provably non-contributory (i.e., features whose removal does not change the model output). A faithful attribution should assign zero relevance to such features. The metric evaluates this by performing a forward pass for each feature (or a downsampled subset thereof) to identify non-contributory features, then measuring the attribution mass on those features. Lower is better.

Non-Sensitivity is computationally intensive, as it requires  $O(d)$  forward passes, where  $d$  is the input dimensionality. In practice, the input is downsampled (e.g., to  $56 \times 56$ ) to make the computation tractable [9]. Due to this cost, Non-Sensitivity is typically excluded from routine evaluations and reserved for targeted diagnostics.

## 2.4. Ensemble XAI: Theoretical Foundations

Given the trade-offs between these metrics - where Gradient methods often excel in Faithfulness but fail in Robustness, and LIME excels in Local Interpretability but fails in Stability - single-method reliance is increasingly viewed as insufficient.

Ensembling in XAI draws inspiration from predictive ensembles (e.g., Random Forests), operating on the hypothesis that aggregating uncorrelated estimators reduces variance and bias [11].

$$E_{\text{ensemble}} = \mathcal{A}(\mathcal{N}(\Phi_1(x)), \mathcal{N}(\Phi_2(x)), \dots, \mathcal{N}(\Phi_n(x))) \quad (2.15)$$

where  $\mathcal{N}$  is a normalization function and  $\mathcal{A}$  is an aggregation function.

Hryniewska-Guzik et al. identify that the primary challenge in XAI ensembling is the heterogeneous scale of attributions (e.g., SHAP values are additive log-odds, while Grad-CAM is feature activation) [11]. They propose **NormEnsembleXAI**, which utilizes robust normalization techniques such as Second Moment Scaling to align these diverse explanations before aggregation via Mean, Max, or Minimum functions. Subsequent work strengthens the ensembling premise: Decker et al. [27] show that an *optimized* convex combination of attribution methods provably improves explanation quality over any single constituent, replacing NormEnsembleXAI’s fixed combinators with learned weights. This thesis builds upon the NormEnsembleXAI framework to evaluate whether ensembling can simultaneously optimize the conflicting objectives of Faithfulness and Robustness.

It is important to distinguish two levels of aggregation that appear in this thesis. *Attribution ensembling*, as performed by NormEnsembleXAI, combines multiple attribution maps  $\Phi_1(x), \dots, \Phi_n(x)$  into a single consolidated explanation  $E_{\text{ensemble}}$  (a third, orthogonal axis—aggregating explanations across *models* rather than across methods—has also been explored [28]). *Metric aggregation*, which constitutes this thesis’s novel contribution, operates at a different level: it combines multiple evaluation metric scores into a single effectiveness index  $E(\Phi)$  that characterizes how well a given FAE method performs across all quality dimensions. Chapter 3 develops the metric-aggregation layer on top of the NormEnsembleXAI normalization framework, treating both individual and ensembled attributions as inputs to the same multi-criteria evaluation pipeline.

## 2.5. Multi-Metric Aggregation and Evaluation Pipelines

The taxonomy presented in Section 2.3 provides a principled vocabulary for assessing individual properties of explanations. In practice, however, no single metric suffices to characterize an FAE method: Faithfulness captures fidelity to the model, Robustness captures stability under perturbation, and Complexity captures cognitive tractability—and these axes are neither independent nor unidirectional. A method may score favorably on one while deteriorating on another, producing rankings that are sensitive to the choice of metric [10, 14]. This section addresses three questions that arise once multiple metrics enter the evaluation: (i) whether the available metrics are mutually informative or merely redundant, (ii) how to validate the metrics themselves, and (iii) how to aggregate non-redundant, validated metrics into a coherent scalar assessment suitable for method comparison.

### 2.5.1. The Metric Redundancy Problem

The proliferation of evaluation metrics within libraries such as Quantus [9] raises a concern that has attracted growing attention: the superset of available metrics contains substantial redundancy, which inflates the apparent rigor of an evaluation without adding information. Formally, given a set of  $K$  metrics  $\{M_1, M_2, \dots, M_K\}$  applied to an explanation  $\Phi(f, x)$ , one can compute the empirical correlation matrix

across a test set  $\mathcal{D}$ :

$$\rho_{ij} = \text{corr}_{x \sim \mathcal{D}}(M_i(\Phi, f, x), M_j(\Phi, f, x)).$$

Tomsett et al. [15] were among the first to quantify this empirically, reporting that metrics within the same category—Faithfulness in particular—exhibit surprisingly low pairwise correlation when applied to image classifiers, despite purporting to measure the same underlying property. This finding has since been replicated across modalities: Bommer et al. [13] observed metric-dependent rankings of XAI methods in a climate-science benchmark, and Klein et al. [29] demonstrated systematic rank reversals between faithfulness operationalizations on a large-scale computer vision benchmark. At the level of individual metrics, Barr et al. [30] document an analogous disagreement problem among faithfulness metrics themselves—the metric-level counterpart of the explainer-level disagreement of Krishna et al. [10]. Pawlicki et al. [31] argue that this proliferation is counterproductive: many widely-used metrics either duplicate information already captured by simpler alternatives or are ineffective by their own operational definition, and recommend a deliberate reduction to a representative subset.

Two practical consequences follow. First, reporting scores across all metrics in a category overstates evidence: a method that wins on five correlated metrics has, in effect, won once. Second, aggregation schemes that treat metrics as independent voters (e.g., averaging ranks across all available faithfulness metrics) are biased toward whichever operational family is most heavily represented in the chosen library. A principled evaluation must therefore begin with a redundancy analysis and proceed with a reduced, justified subset. Meta-reviews of the metric landscape reach the same conclusion from the reporting side: Gipiškis and Kurasova [32] find that correlations between related metrics are rarely reported at all, and propose evaluation cards that make them a required reporting field.

### 2.5.2. Meta-Evaluation: Validating the Validators

The redundancy problem reveals that metrics disagree with each other; a deeper problem is that it is not always clear which metric is *correct*. Since no ground-truth explanation exists against which to calibrate a metric, the standard evaluation paradigm lacks the reference signal used elsewhere in supervised machine learning. Hedström et al. [14] formalize this as the *meta-evaluation problem*: how does one evaluate a metric that is itself supposed to evaluate an explanation? Their proposed framework, MetaQuantus, subjects each metric to two stress tests grounded in plausible failure modes:

- **Noise Resilience (NR)**: under controlled, semantically meaningless perturbations of model inputs or weights, a reliable metric should yield stable scores. High score variance under noise indicates that the metric is responding to spurious signal.
- **Adversarial Reactivity (AR)**: under controlled degradation of the explanation itself (e.g., partial randomization), a reliable metric should show a monotonic drop in score. A metric that remains unchanged under explanation degradation is not tracking explanation quality.

A metric that performs poorly on either test is deprioritized as an unreliable signal, regardless of its theoretical motivation. (This thesis implements *MetaQuantus-inspired* estimators of these two properties rather than the published estimators; the operationalization is specified in Chapter 3.) The concern is not merely theoretical: evaluating metrics against transparent models whose ground-truth explanations are known, Miró-Nicolau, Capó, and Moyà-Alcover [33] conclude that current fidelity metrics are not yet reliable enough for real-world scenarios, which underscores the need to validate a metric before trusting its verdict, and Fresz, Lörcher, and Huber [34] likewise treat the reliability of the evaluation metrics themselves—their objectivity and consistency in the psychometric sense—as a first-class requirement for trustworthy XAI evaluation. This meta-evaluation layer is essential for the framework proposed in this thesis, as it provides a principled criterion for the *weighting* of metrics in the aggregation stage of Chapter 3, complementing the correlation-based redundancy analysis of Section 2.5.1.

### 2.5.3. Normalization and Aggregation Strategies

Once a non-redundant, meta-validated subset  $\{M_1, \dots, M_K\}$  is selected, the individual scores must be combined into a scalar that supports method comparison. Two technical obstacles arise: heterogeneous scales and heterogeneous directions. A Faithfulness Correlation score lies in  $[-1, 1]$ ; a Max-Sensitivity score is unbounded above and lower-is-better; a Sparseness score is in  $[0, 1]$  and higher-is-better. Direct averaging is meaningless without rectification.

Let  $M_k(\Phi)$  denote the raw score of metric  $k$  for explanation  $\Phi$ , and let  $s_k \in \{+1, -1\}$  encode the direction convention (+1 for higher-is-better). A normalized score  $\tilde{M}_k$  is obtained via

$$\tilde{M}_k(\Phi) = s_k \cdot \psi_k(M_k(\Phi)),$$

where  $\psi_k$  is a scale-rectifying transformation. Three choices dominate the literature:

1. **Min–max normalization:**  $\psi_k(x) = (x - \min_k) / (\max_k - \min_k)$ , mapping scores to  $[0, 1]$ . Sensitive to outliers.
2.  **$z$ -score normalization:**  $\psi_k(x) = (x - \mu_k) / \sigma_k$ , centering on the mean with unit variance.
3. **Second Moment Scaling:**  $\psi_k(x) = x / \sqrt{\mathbb{E}[x^2]}$ , employed by Hryniewska-Guzik, Sawicki, and Biecek [11] in the NormEnsembleXAI framework to align explanations of heterogeneous origin (e.g., SHAP’s log-odds contributions vs. Grad-CAM’s activation magnitudes) prior to ensemble aggregation.

Once normalized, aggregation proceeds via a scalar combinator  $\mathcal{A}$ , yielding the effectiveness index

$$E(\Phi) = \mathcal{A}(\tilde{M}_1(\Phi), \tilde{M}_2(\Phi), \dots, \tilde{M}_K(\Phi)).$$

Common choices for  $\mathcal{A}$  include the weighted arithmetic mean  $\sum_k w_k \tilde{M}_k$ , the minimum  $\min_k \tilde{M}_k$  (a conservative operator favoring methods acceptable on all criteria), and the geometric mean (penalizing methods that fail severely on any single axis). The minimum operator corresponds to evaluating a method

by its worst dimension and has a natural interpretation in safety-critical domains: a method that is faithful but unstable is ineligible regardless of how high its faithfulness score is. Beyond such fixed combinators, recent work frames the combination step as a multi-criteria decision-making (MCDM) problem: Chatterjee, Colombo, and Raimundo [35] weight competing quality metrics with TOPSIS-style operators and aggregate explainer *rankings* rather than scores, offering a rank-based alternative to the score-based weighted mean adopted in this thesis.

#### 2.5.4. Evaluation Pipelines and the Autoweighted System

The framework introduced above—(i) redundancy-aware metric selection, (ii) meta-validation, (iii) direction rectification and scale normalization, (iv) aggregation—constitutes an *evaluation pipeline*. The pipeline abstraction is significant because it isolates configuration decisions (which metrics, which normalization, which aggregator, which weights) from the XAI method under evaluation, enabling reproducible, principled comparison across studies.

A key design choice within this pipeline is the weight vector  $\mathbf{w} = (w_1, \dots, w_K)$ . Fixed equal weighting assumes all retained criteria contribute equally—a strong assumption that the redundancy analysis alone does not validate. Hryniewska-Guzik, Sawicki, and Biecek [11] introduce the *Autoweighted* mechanism, a data-adaptive weighting in which each component’s weight is derived from the statistics of its own scores rather than set by hand, so that dominant, trustworthy signals receive higher weight without manual tuning. This thesis adapts the Autoweighted idea to the metric-aggregation layer by weighting each metric by the inverse coefficient of variation of its scores across the test set—metrics whose scores are internally consistent are trusted more than metrics whose scores are noisy—with the exact formulation given in Chapter 3 (Equation (3.6)). Crucially, the thesis keeps this Autoweighted scheme intact and *separately* formulates a MetaQuantus-discounted extension (Equation (3.8)), in which each metric’s Autoweighted coefficient is attenuated by its meta-evaluation reliability (Noise Resilience and Adversarial Reactivity from Hedström et al. [14]). Keeping the two schemes distinct is what allows the experiments of Chapter 4 to isolate the effect of the reliability discount on the resulting FAE ranking.

Recent work has emphasized that evaluation pipelines are themselves a compliance instrument. Seth and Sankarapu [36] argue that standardized, reproducible pipelines—rather than ad-hoc metric selection—are a prerequisite for operationalizing the transparency requirements of the EU AI Act, aligning the technical contribution of this thesis with the regulatory motivation established in Section 1.1.

## 3. Methodology

This chapter presents the evaluation framework developed in this thesis. Section 3.1 defines the evaluation objective and the unit of analysis. Sections 3.2–3.4 describe the dataset, predictive models, and feature attribution methods that constitute the experimental infrastructure. Section 3.5 specifies the twelve evaluation metrics organized across six Quantus categories. Sections 3.6–3.7 detail the score normalization and aggregation procedures that form the core methodological contribution of this work (Contributions 1–3, per Section 1.2). Finally, Section 3.8 describes the end-to-end evaluation pipeline and its software implementation.

### 3.1. Evaluation Setup

The objective of this evaluation framework is to rank Feature Attribution Explanation (FAE) methods by an aggregated effectiveness index  $E(\Phi)$  that summarizes performance across six metric categories: Faithfulness, Robustness, Localization, Complexity, Randomization, and Axiomatic.

The atomic unit of evaluation is a *per-(model, image, FAE method) attribution triple*. For a classifier  $f$ , an input image  $x$ , and an explanation method  $\Phi$ , the triple  $(f, x, \Phi)$  yields an attribution map  $\Phi(f, x) \in \mathbb{R}^{C \times H \times W}$ , where  $C, H, W$  denote the channel, height, and width dimensions of the input.

For each triple, the **target class** used for attribution is the model’s top-1 prediction  $\hat{y} = \arg\max_c f(x)_c$ , not the ground-truth label. This choice is deliberate: evaluation metrics assess explanation fidelity to *the model’s decision*, not to a human-supplied label. An explanation that faithfully reflects the features driving the model’s prediction is informative regardless of whether that prediction is correct [9].

### 3.2. Dataset

The experiments use the ISIC 2017 Challenge dataset (Task 1 — Lesion Segmentation), publicly available under a CC-BY-NC license from [challenge.isic-archive.com](http://challenge.isic-archive.com). The dataset comprises dermoscopic images distributed across three diagnostic categories: melanoma, nevus, and seborrheic keratosis.

- **Split sizes:** 2,000 training images, 150 validation images, 600 test images.

- **Segmentation masks:** Each image is accompanied by an expert-annotated binary mask delineating the lesion boundary. These masks enable the computation of Localization metrics (Pointing Game, Relevance Mass Accuracy) without additional labeling effort—a key advantage over datasets that provide only image-level labels.
- **Preprocessing:** All images are resized to  $224 \times 224$  pixels and normalized using ImageNet channel-wise mean and standard deviation ( $\mu = [0.485, 0.456, 0.406]$ ,  $\sigma = [0.229, 0.224, 0.225]$ ).

The vertical-slice validation (Section 3.8) uses a fixed subset of 12 images drawn from the test split to verify the end-to-end plumbing at low computational cost; the full experiment scales the identical pipeline to all 600 test images. The split sizes above (2,000 / 150 / 600) are those of the official ISIC 2017 Task 1 release and are held fixed for all experiments to preserve comparability with the challenge protocol.

### 3.3. Predictive Models

Two convolutional neural network architectures serve as the predictors whose decisions are explained. They are not a contribution of this thesis; they exist solely to produce the attributions that the evaluation framework then scores.

#### 3.3.1. ResNet-18

ResNet-18 is a residual network with 18 layers organized in four residual blocks with skip connections [37]. The final fully connected layer is replaced with `Linear(512, 3)` to match the three-class ISIC 2017 classification task.

#### 3.3.2. SqueezeNet 1.1

SqueezeNet 1.1 is a compact architecture that achieves AlexNet-level accuracy with  $\sim 50\times$  fewer parameters, using Fire modules that combine  $1 \times 1$  squeeze convolutions with mixed  $1 \times 1$  and  $3 \times 3$  expand convolutions [38]. The final classifier layer is replaced with `Conv2d(512, 3, 1, 1)` for three-class output.

#### 3.3.3. Training Protocol

Both models are trained with the following configuration:

- **Epochs:** 25
- **Optimizer:** SGD with momentum ( $\text{lr} = 0.001$ ,  $\text{momentum} = 0.9$ )
- **Scheduler:** StepLR (step size 7,  $\gamma = 0.1$ )

- **Loss:** Class-weighted CrossEntropyLoss to address class imbalance in the ISIC 2017 dataset
- **Random seed:** 42 (set for `torch`, `numpy`, and `random`)

Best validation accuracies: ResNet-18 achieves 75.3%, SqueezeNet achieves 74.0%. Trained weights are saved to disk with SHA-256 checksums documented in `weights/README.md`.

## 3.4. Feature Attribution Methods

Seven attribution methods are evaluated, spanning gradient-based (white-box) and perturbation-based (black-box) families. All methods are implemented using the Captum library, ensuring consistent interfaces and reproducible gradient computation.

### 3.4.1. Gradient-based Methods

#### 3.4.1.1. Saliency (Vanilla Gradient)

Computes the absolute gradient of the class score with respect to the input image (Chapter 2). Computationally efficient but prone to noisy, non-class-discriminative maps due to shattered gradients.

#### 3.4.1.2. Integrated Gradients

Accumulates gradients along a linear path from a black baseline ( $x' = 0$ ) to the actual input  $x$  using  $n\_steps = 50$  interpolation points [19]. Satisfies the Completeness axiom by construction.

#### 3.4.1.3. Grad-CAM

Gradient-weighted Class Activation Mapping applied to the last convolutional layer of the network [3]. Produces a coarse localization map at the resolution of the final feature map, which is then upsampled to input resolution via bilinear interpolation.

#### 3.4.1.4. Guided Backpropagation

Propagates gradients through only the positive activations at each ReLU layer, producing sharper maps than vanilla gradients but at the risk of acting as an edge detector independent of model parameters [39].

#### 3.4.1.5. DeepLift

Computes attribution by comparing each neuron’s activation to a reference activation produced by a black baseline input ( $x' = 0$ ). Decomposes the output difference  $f(x) - f(x')$  into individual feature contributions [40]. Because the torchvision `BasicBlock` (ResNet) and `Fire` (SqueezeNet) modules reuse a single in-place ReLU across multiple call sites, the implementation operates on a deep copy of the model in which shared ReLU modules are replaced by unique non-in-place instances, as required by the Captum DeepLift backend.

### 3.4.1.6. Layer-wise Relevance Propagation (LRP)

Redistributes the model output backward through the network according to propagation rules that preserve the total relevance at each layer [22]. Captum’s default LRP rules are applied; both base classifiers are composed exclusively of layer types supported by these rules.

## 3.4.2. Perturbation-based Methods

### 3.4.2.1. Occlusion

A sliding window of size (3, 15, 15) with stride (3, 8, 8) is passed over the input, replacing occluded regions with a black baseline ( $x' = 0$ ). The change in model output for each window position yields a perturbation sensitivity map [41]. Occlusion is the sole perturbation-based method retained in the final evaluation set.

### 3.4.3. Excluded Candidates

LIME and KernelSHAP were initially considered as additional perturbation-based (black-box) methods, alongside Occlusion. They were excluded from the final evaluation set after compatibility testing on the vertical slice. The Robustness metrics (Max-Sensitivity and Avg-Sensitivity) and the Randomization metrics re-invoke the attribution method on perturbed inputs or randomized models through a Quantus `explain_func` callback; the sampling-based LIME and KernelSHAP estimators are both prohibitively expensive to re-run for every perturbation sample and produce attribution maps whose stochastic surrogate fitting makes the resulting sensitivity scores unstable and difficult to interpret. Occlusion provides a deterministic, Captum-native perturbation baseline that integrates cleanly with these inner re-computations, so it alone represents the perturbation family. The final evaluation set is therefore the seven methods above: six gradient/back-propagation methods (Saliency, Integrated Gradients, Grad-CAM, Guided Backpropagation, DeepLift, and LRP) and one perturbation method (Occlusion).

### 3.4.4. Attribution Caching

Attributions are cached to disk, keyed by a tuple of (model architecture, FAE method, image identifier, target class, hyperparameters hash). This avoids redundant recomputation when the same attribution is needed by multiple metrics or across repeated pipeline runs.

The cache is **bypassed** for inner re-computations performed by robustness and randomization metrics—specifically Max-Sensitivity, Avg-Sensitivity, and MPRT—because these metrics intentionally perturb inputs or model weights and therefore produce legitimately different attributions on each invocation. Caching such perturbed attributions would violate the metric’s operational semantics.

### 3.5. Metric Computation

Twelve metrics across six Quantus categories are computed for each attribution triple. Table 3.1 summarizes their directionality and key hyperparameters.

**Tab. 3.1.** Evaluation metrics: category, direction, and hyperparameters.

| Category      | Metric                      | Direction | Key hyperparameters                                    |
|---------------|-----------------------------|-----------|--------------------------------------------------------|
| Faithfulness  | FaithfulnessCorrelation     | ↑         | nr_runs=100, subset_size=224                           |
| Faithfulness  | PixelFlipping               | ↓         | features_in_step=224 (AUC of MoRF curve)               |
| Robustness    | MaxSensitivity              | ↓         | nr_samples=10, lower_bound=0.2                         |
| Robustness    | AvgSensitivity              | ↓         | nr_samples=10, lower_bound=0.2                         |
| Localization  | RelevanceMassAccuracy       | ↑         | requires mask                                          |
| Localization  | PointingGame                | ↑         | requires mask                                          |
| Complexity    | Sparseness                  | ↑         | Gini-like coefficient                                  |
| Complexity    | Complexity                  | ↓         | Shannon entropy                                        |
| Randomization | ModelParameterRandomisation | ↓         | layer_order=top_down; excluded on ISIC <sup>†</sup>    |
| Randomization | RandomLogit                 | ↓         | num_classes=3                                          |
| Axiomatic     | Completeness                | ↑         | black baseline; IG/DeepLift/LRP only                   |
| Axiomatic     | NonSensitivity              | ↓         | downsampled to 56×56; excluded by default <sup>‡</sup> |

The schema retains all twelve metric slots for every attribution triple so that the result tables have a stable shape, but two of them are excluded from computation as implementation-driven decisions and recorded as NaN (subsequently dropped from aggregation rather than imputed):

† **ModelParameterRandomisation** is disabled on ISIC 2017. Under Quantus 0.6.0 its implementation raises an `AssertionError` on every (model, image) sample in this dataset and additionally emits a per-call deprecation notice, so it contributes only NaN while flooding the log and wasting compute. The exclusion is consistent with the literature: the metric’s own authors have since documented methodological flaws in the original MPRT formulation and proposed repaired variants (Smooth-MPRT and Efficient-MPRT) [42]. The Randomization category remains represented by RandomLogit, which exercises the same sanity-check principle (an explanation should change when the target logit is replaced), so the category is not lost.

‡ **NonSensitivity** is excluded by default. A Quantus shape bug forces `features_in_step = 1`, turning the metric into one forward pass per feature; at  $224 \times 224 \times 3$  this is roughly 150,000 passes

per image. Even with the input and attribution downsampled to  $56 \times 56$  (about 9,000 passes,  $\sim 34$  s per image) the cost is prohibitive at the scale of the full experiment. It can be enabled through an `include_non_sensitivity` flag for targeted runs.

With these two exclusions, ten metrics are computed per attribution triple in the default ISIC 2017 configuration (FaithfulnessCorrelation, PixelFlipping, MaxSensitivity, AvgSensitivity, RelevanceMassAccuracy, PointingGame, Sparseness, Complexity, RandomLogit, and Completeness, with Completeness itself defined only for the three completeness-claiming methods). All six Quantus categories remain covered.

**3.5.0.0.1 Faithfulness.** FaithfulnessCorrelation and PixelFlipping both quantify how well the attribution map identifies features that the model relies on for its prediction. FaithfulnessCorrelation uses random subset masking with 100 runs to estimate the Pearson correlation between attribution magnitude and output change; higher is better. PixelFlipping removes features in descending order of attributed importance and reports the area under the degradation curve of model outputs; a faithful attribution collapses the prediction early, so *lower* is better [9].

**3.5.0.0.2 Robustness.** MaxSensitivity and AvgSensitivity evaluate explanation stability under small uniform-noise perturbations of the input (the Quantus implementation draws uniform noise with magnitude bounded by `lower_bound`). Both use 10 perturbation samples with a perturbation magnitude lower bound of 0.2. MaxSensitivity reports the worst-case sensitivity, while AvgSensitivity reports the average. Lower is better for both [9].

**3.5.0.0.3 Localization.** RelevanceMassAccuracy and PointingGame evaluate spatial agreement between the attribution map and the expert-annotated lesion mask. Both are higher-is-better and require segmentation masks at inference time [9].

**3.5.0.0.4 Complexity.** Sparseness (Gini-like, higher-is-better) and Complexity (Shannon entropy, lower-is-better) assess the conciseness of the attribution distribution. These metrics are model-agnostic and computed directly from the attribution map [9].

**3.5.0.0.5 Randomization.** ModelParameterRandomisation (MPRT) progressively randomizes model weights top-down and measures correlation decay. RandomLogit replaces the target class and measures explanation change. Lower correlation is better for both, indicating that the explanation genuinely depends on learned parameters and the specified target class [9]. As noted above, MPRT is disabled on ISIC 2017 because of a Quantus 0.6.0 incompatibility, leaving RandomLogit as the operative Randomization metric.

**3.5.0.0.6 Axiomatic.** Completeness measures the absolute deviation from the completeness axiom ( $|\sum_i E_i - \Delta f| \approx 0$ ). It returns NaN for methods outside {Integrated Gradients, DeepLift, LRP}; these NaN values are excluded from aggregation rather than imputed. NonSensitivity measures attribution mass on provably non-contributory features. Due to its  $O(d)$  forward-pass cost, the input is downsampled to  $56 \times 56$ , and the metric is excluded from all runs by default [9].

### 3.5.1. Metric Redundancy Pruning

The reduced metric set  $M^*$  (Contribution 1) is derived per model in three steps.

*Variance pre-screen.* Within each model group, a metric is removed before correlation analysis if its non-NaN fraction falls below 0.20 or the standard deviation of its scores falls below  $10^{-6}$  (a constant metric carries no ranking signal; a mostly-NaN metric yields unstable correlation estimates).

*Correlation matrix.* For the surviving metrics, the Spearman rank-correlation matrix is computed on the wide table whose observations are (image, FAE-method) pairs — i.e. correlations are pooled across attribution methods as well as images, so “redundant” means that two metrics order the full population of explanations alike, not merely that they agree for one method. Spearman is used rather than Pearson because the raw metric scales are heterogeneous and partly unbounded.

*Greedy pruning (Decision D4).* While any pair of surviving metrics has  $|\rho| > 0.85$ , the member of the pair with the *higher* mean absolute correlation to the rest of the surviving set is dropped (the retained member is the one carrying more independent signal); the procedure iterates until no pair exceeds the threshold. The threshold 0.85 follows the convention of treating  $|\rho|$  above  $\sim 0.85$  as operational collinearity; Chapter 5 discusses its sensitivity. The procedure is run per model, and the reported  $M^*$  is the cross-model intersection.

## 3.6. Score Normalization

The twelve metrics operate on heterogeneous scales (e.g., FaithfulnessCorrelation  $\in [-1, 1]$ , MaxSensitivity  $\in [0, \infty)$ , Sparseness  $\in [0, 1]$ ) and have heterogeneous directionalities (some higher-is-better, some lower-is-better). Aggregation requires rectification and rescaling.

### 3.6.1. Direction Rectification

Each metric  $k$  is assigned a direction sign  $s_k \in \{+1, -1\}$ , where  $s_k = +1$  for higher-is-better metrics and  $s_k = -1$  for lower-is-better metrics. The rectified score is:

$$\hat{M}_k(\Phi) = s_k \cdot M_k(\Phi), \quad (3.1)$$

so that a higher rectified score always indicates better performance.

### 3.6.2. Second Moment Scaling

The default scale-rectifying transformation is Second Moment Scaling, adopted from the NormEnsembleXAI framework (Decision D1) [11]:

$$\tilde{M}_k(\Phi) = \frac{\hat{M}_k(\Phi)}{\sqrt{\mathbb{E}[\hat{M}_k^2]}}, \quad (3.2)$$

where the expectation is computed within each (model, metric) group independently. This transformation does not require knowledge of global extremes, is robust to outliers (unlike min–max normalization), and preserves the sign structure of the rectified scores.

Two alternative normalizations are implemented for ablation studies:

- **z-score normalization:**  $\psi_k(x) = (x - \mu_k)/\sigma_k$ , centering on the group mean with unit variance.
- **Min–max normalization:**  $\psi_k(x) = (x - \min_k)/(\max_k - \min_k)$ , mapping scores to  $[0, 1]$ . Sensitive to outliers and requires knowing the global extremes.

## 3.7. Score Aggregation

Once normalized, the per-metric scores are aggregated into a single effectiveness index  $E(\Phi)$  per (model, image, FAE method). Three aggregation operators are implemented (Decision D2):

### 1. Weighted mean (default, recommended):

$$E(\Phi) = \sum_{k=1}^K w_k \tilde{M}_k(\Phi), \quad (3.3)$$

where  $w_k \geq 0$  and  $\sum_k w_k = 1$ .

### 2. Minimum (conservative):

$$E(\Phi) = \min_{k=1, \dots, K} \tilde{M}_k(\Phi). \quad (3.4)$$

Evaluates a method by its worst dimension. In safety-critical domains, a method that is faithful but unstable is ineligible regardless of how high its faithfulness score is.

### 3. Geometric mean:

$$E(\Phi) = \exp \left( \sum_{k=1}^K w_k \ln(\tilde{M}'_k(\Phi) + \varepsilon) \right), \quad (3.5)$$

where  $\varepsilon > 0$  is a small constant to avoid  $\ln(0)$  and  $\tilde{M}'_k = \tilde{M}_k - \min_j \tilde{M}_j(\cdot)$  shifts the rectified scores to non-negative support first (rectified scores of lower-is-better metrics are routinely negative, so the logarithm is otherwise undefined). Penalizes methods with a near-zero score on any single axis.

The default weight vector uses uniform weights:  $w_k = 1/K$  for  $K$  retained metrics after the redundancy and meta-evaluation filtering whose procedure is specified in Chapter 4 (the underlying redundancy and meta-evaluation concepts are reviewed in Chapter 2).

### 3.7.1. Autoweighted Extension

The Autoweighted mechanism, adopted from Hryniewska-Guzik, Sawicki, and Biecek [11], derives a weight for each metric from the dispersion of its scores across the test set: metrics whose scores are internally consistent are trusted more than metrics whose scores are noisy. Concretely, for each (model) group and each metric  $k$ , let  $cv_k = \text{std}(M_k)/|\text{mean}(M_k)|$  be the coefficient of variation of metric  $k$ 's raw scores over the test images. The agreement weight is the inverse coefficient of variation, normalised to a convex combination:

$$a_k = \frac{1}{cv_k + \varepsilon}, \quad w_k^{\text{AW}} = \frac{a_k}{\sum_{j \in M^*} a_j}, \quad (3.6)$$

where  $\varepsilon = 10^{-8}$  provides numerical stability and all statistics are NaN-aware. A metric with a low coefficient of variation (stable across images relative to its mean) receives a high weight; a noisy metric receives a small one. Metrics whose scores are all-NaN in a group receive  $a_k = 0$ , and the remaining weights renormalise so that  $\sum_k w_k^{\text{AW}} = 1$ . The statistics are pooled across all FAE methods and test images within a model group. The coefficient of variation is scale-invariant, so Autoweighted yields the same result whether computed on raw or Second-Moment-scaled scores (this invariance does not extend to the  $z$ -score alternative, whose centring removes the mean the CV depends on).

Two structural caveats of inverse-CV weighting are acknowledged as modelling assumptions rather than hidden. First, the CV is *location-sensitive*: for a metric whose scores straddle zero — Faithfulness-Correlation, with range  $[-1, 1]$ , is the case in point —  $|\text{mean}(M_k)|$  can be small for reasons unrelated to noise, which inflates the CV and suppresses the weight as an artefact of where the scale's zero sits. Second, consistency is not discriminative power: a degenerate metric returning a near-constant score would receive maximal weight while carrying no ranking signal (the variance pre-screen of Section 3.5.1 removes the limiting case, but the gradient of the pathology remains). Both caveats motivate testing, rather than assuming, whether the reliability discount of the next subsection changes the induced ranking.

### 3.7.2. MetaQuantus-discounted Weighting (Contribution 2)

The reliability-discounted weighting scheme of this thesis (Decision D3) combines two signals that are conventionally treated in isolation: how internally consistent a metric's scores are (the Autoweighted signal above), and how reliable that metric is when probed by meta-evaluation. The derivation proceeds in three steps.

**3.7.2.0.1 Step 1 — Autoweighted agreement weight.** For each retained metric  $k \in M^*$ , the Autoweighted coefficient  $w_k^{\text{AW}}$  is computed per Equation (3.6).

**3.7.2.0.2 Step 2 — Reliability discount.** Each metric is assigned a reliability coefficient  $\lambda_k \in [0, 1]$  obtained from the meta-evaluation stage (Section 2.5.2 of Chapter 2). MetaQuantus produces two scores per metric: a Noise Resilience score  $\text{NR}_k$  and an Adversarial Reactivity score  $\text{AR}_k$ . These are this thesis's MetaQuantus-inspired estimators of the two reliability properties, not the published IAC/IEC

consistency estimators:  $\text{NR}_k = 1/(1 + \text{CV})$ , where  $\text{CV}$  is the coefficient of variation of the metric's score across five noise seeds, and  $\text{AR}_k = |\rho_S|$ , the absolute Spearman correlation between five explanation-degradation levels and the mean metric score at each level. Both are computed on a fixed, seed-determined 64-image random sample of the test split per (model, FAE) cell; the extended-AR completion for the two explanation-recomputing metrics used a fixed 12-image split-order sample, distinct from the 64-image draw, so the combined  $\lambda_k$  for those two metrics mixes the two samples. The combined reliability is their equal-weight mean,

$$\lambda_k = \frac{1}{2}(\text{NR}_k + \text{AR}_k), \quad (3.7)$$

reported per (model, FAE, metric) in `results/meta_evaluation_reliability.csv` and averaged over the (model, FAE) cells in which the metric is defined. Because  $|\rho_S|$  over five levels has a granularity of  $\sim 0.1$  and is direction-agnostic, reliability values are reported to two decimals and the direction of each cell's reactivity is audited in Chapter 4.

The AR test requires one protocol extension. MetaQuantus degrades the explanation by zeroing a growing fraction of the attribution array, but the metrics that re-compute attributions internally through `explain_func` (the sensitivity metrics and `RandomLogit`) ignore that array, so the degradation would never reach them and their level scores would be constant. For those metrics the degradation is therefore additionally injected into the explanation function itself: at degradation fraction  $f$ , each invocation of the wrapped explainer zeroes a fresh, independently drawn random fraction  $f$  of its output, so the metric's internal re-explanations lose signal at the same rate as the explicit attribution. The degradation is capped at  $f = 0.9$  because a fully zeroed attribution is rejected by the metric implementations. With this extension, every computable metric carries a measured combined reliability.

Where `combined_reliability` is undefined for a metric that nonetheless enters an aggregation — a configuration safeguard: in the reported run every  $M^*$  metric carries a measured reliability, so the hierarchy is not exercised —  $\lambda_k$  is resolved by an explicit fallback hierarchy: (i) the Noise-Resilience-only score, where NR alone is available; (ii) the mean reliability of the other valid metrics in the same Quantus category; (iii) the global mean reliability across the group; and, as a last resort, (iv) no discount ( $\lambda_k = 1$ ). This fallback never zeroes a metric out, which would silently drop it from the ensemble.

**3.7.2.0.3 Step 3 — Discounted normalisation.** The Autoweighted coefficient is attenuated by the reliability factor and renormalised to a convex combination:

$$w_k = \frac{w_k^{\text{AW}} \lambda_k}{\sum_{j \in M^*} w_j^{\text{AW}} \lambda_j}, \quad w_k \geq 0, \quad \sum_{k \in M^*} w_k = 1. \quad (3.8)$$

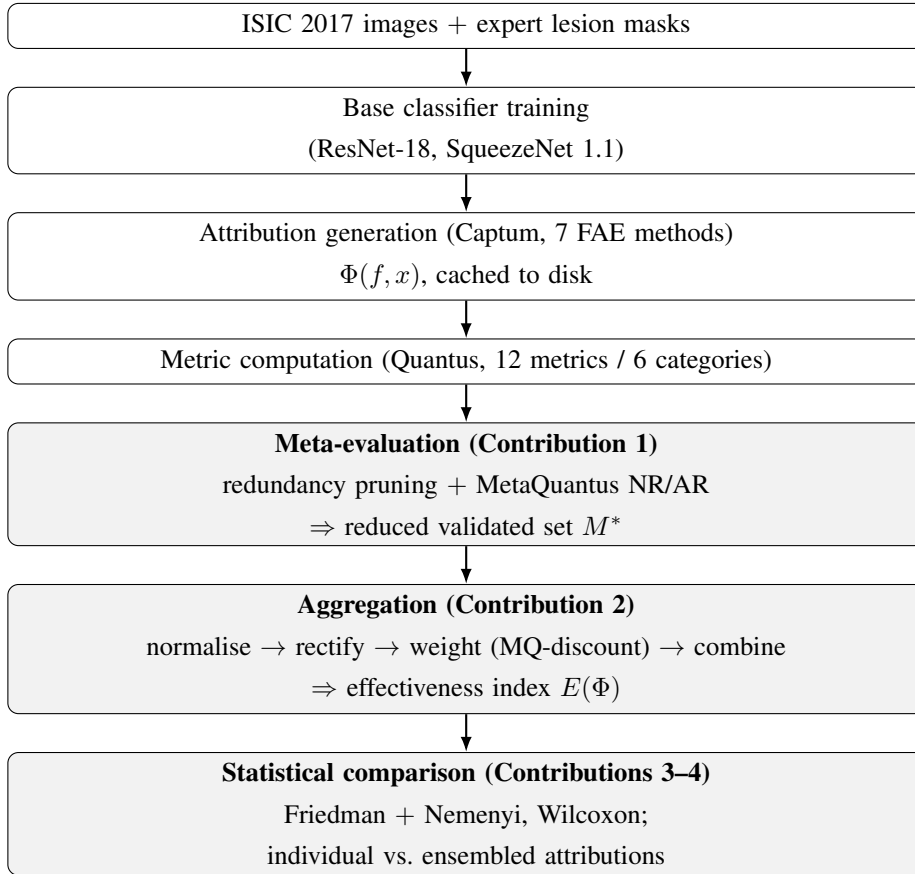
Equation (3.8) is the MetaQuantus-discounted Autoweighted operator. It reduces to the plain Autoweighted scheme of Hryniewska-Guzik, Sawicki, and Biecek [11] when every metric is perfectly reliable ( $\lambda_k \equiv 1$ ), and it drives  $w_k \rightarrow 0$  for any metric that either is internally noisy (small  $w_k^{\text{AW}}$ ) or fails meta-evaluation (small  $\lambda_k$ ). The three weighting schemes—uniform, Autoweighted, and MQ-discounted—are reported side by side in Chapter 4 so that their effect on the final method ranking can be assessed directly.

### 3.8. The Evaluation Pipeline

The end-to-end evaluation pipeline integrates all components described in this chapter into a single reproducible data flow:

1. **Input:** ISIC 2017 test images and associated binary lesion masks.
2. **Model inference:** For each image  $x$ , the classifier  $f$  produces a prediction  $\hat{y} = \operatorname{argmax}_c f(x)_c$ , which serves as the target class for attribution.
3. **Attribution generation:** Each FAE method  $\Phi$  computes an attribution map  $\Phi(f, x) \in \mathbb{R}^{3 \times 224 \times 224}$ . Results are cached to disk (Section 3.4.4).
4. **Metric computation:** The twelve metrics from Table 3.1 are applied to each attribution map, producing a vector of raw scores per (model, image, FAE method).
5. **Normalization:** Direction rectification and Second Moment Scaling (Section 3.6) transform raw scores into a comparable scale.
6. **Aggregation:** The normalized scores are combined into a single effectiveness index  $E(\Phi)$  via the chosen aggregation operator (Section 3.7).
7. **Ranking:** FAE methods are ranked per-model by their mean  $E(\Phi)$  across test images.

Figure 3.1 summarises this flow. The diagram makes explicit the two contribution boundaries: the meta-evaluation block (redundancy + meta-validation) yields the reduced metric set  $M^*$  (Contribution 1), and the aggregation block (normalisation, weighting, combination) yields the effectiveness index  $E(\Phi)$  (Contribution 2), which the statistical-comparison stage (Contributions 3–4) consumes.



**Fig. 3.1.** End-to-end evaluation pipeline. White boxes are experimental infrastructure; shaded boxes are the methodological contributions of this thesis.

### 3.8.1. Implementation

The pipeline entry point is `experiments/vertical_slice.py`. The output is a long-format CSV file with columns `model`, `image_id`, `fae_method`, `metric`, and `score`. The schema retains one row per (model, image, FAE method, metric slot), including the slots recorded as NaN (the excluded metrics and Completeness on non-completeness methods), so that the table shape is independent of which metrics happen to be defined for a given triple. For the 12-image vertical slice this is  $2 \times 7 \times 12 \times 12 = 2,016$  rows (2 models  $\times$  7 FAE methods  $\times$  12 images  $\times$  12 metric slots); scaling the identical schema to all 600 test images yields  $2 \times 7 \times 600 \times 12 = 100,800$  rows.

The software architecture follows the modular layout described in Section 1.2:

- `src/data/` — ISIC 2017 dataset class and download script.
- `src/models/` — Classifier definitions and training loop.
- `src/attributions/` — Attribution generation, disk cache, NormEnsembleXAI wrapper.
- `src/metrics/` — Quantus wrapper, custom metrics, redundancy analysis.

- `src/meta_evaluation/` — MetaQuantus wrapper and metric selection.
- `src/aggregation/` — Normalization, aggregation operators, weighting schemes.
- `src/comparison/` — Wilcoxon, Friedman, and Nemenyi statistical tests.
- `src/pipeline.py` — End-to-end orchestrator.

All experiments are configured via YAML files under `configs/`, with one configuration per experiment variant. Random seeds, model paths, metric hyperparameters, and normalization choices are specified declaratively to ensure full reproducibility.

**3.8.1.0.1 Design decision register.** For ease of reference, the design decisions cited as D1–D6 throughout Chapters 4 and 5 are collected here:

- D1** Second Moment Scaling as the score normalisation, adopted from NormEnsembleXAI (Section 3.6).
- D2** Three aggregation operators—weighted mean, minimum, and geometric mean—with the weighted mean as the default (Section 3.7).
- D3** The MetaQuantus-discounted Autoweighted weighting as the novel weighting scheme under test (Section 3.7.2).
- D4** Greedy redundancy pruning at  $|\rho| > 0.85$  over the pooled (cross-category) correlation matrix, with variance pre-screening (Section 3.5.1); a strictly within-category policy is the considered alternative.
- D5** Friedman + Nemenyi for the omnibus method comparison and Wilcoxon signed-rank for the paired individual-vs-ensemble comparison (Section 3.8).
- D6** NormEnsembleXAI Mean aggregation with Second Moment normalisation as the attribution-ensembling configuration (Section 3.4).



## 4. Experiments and Results

This chapter reports the empirical evaluation of the framework developed in Chapter 3. The experiments are organised to mirror the four research contributions of Section 1.2: Section 4.1 fixes the experimental configuration; Section 4.2 presents the metric redundancy analysis (Contribution 1); Section 4.3 reports the meta-evaluation of metric reliability (Contribution 2); Section 4.4 compares the uniform, Autoweighted, and MetaQuantus-discounted aggregation schemes (Contributions 2–3); Section 4.5 contrasts individual and NormEnsembleXAI-ensembled attributions (Contribution 3); and Section 4.6 establishes statistical significance via Friedman, Nemenyi, and Wilcoxon testing (Contribution 4).

All quantitative tables in this chapter are generated directly from the result artifacts under `results/` by `experiments/render_results_tables.py` and \input-ed here. The scores, rankings, ensemble comparison, and statistical tests are computed over the full 600-image ISIC 2017 test split; the meta-evaluation (Section 4.3) is the deliberate exception, running on a fixed 64-image sample (12 images for the extended-AR completion), as disclosed where its numbers are used. Two further limitations are stated up front and revisited at the point of use: (i) scoring the Adversarial Reactivity of the two  $M^*$  metrics that internally re-compute attributions (*max\_sensitivity*, *random\_logit*) required an extension of the AR protocol—the degradation is injected into the explanation function itself (Section 4.3); with that extension every  $M^*$  metric carries a meta-evaluated combined reliability, and the MQ-discount genuinely reweights the index (mean  $|\Delta| \approx 0.07$  relative to Autoweighted, with a model-dependent effect on the ranking); and (ii) the attribution-ensembling experiment of Section 4.5 ensembles the six gradient-based methods only (Occlusion excluded as an ensemble member for tractability), though it otherwise uses the full 600 images and all seven  $M^*$  metrics. Cells reading *n/a* denote a value that is genuinely undefined (e.g. a single-metric category has no within-category correlation), not a missing computation.

### 4.1. Experimental Setup

#### 4.1.1. Data and Splits

All experiments use the ISIC 2017 Challenge dataset (Task 1) described in Section 3.2: 2,000 training, 150 validation, and 600 test dermoscopic images across three diagnostic classes (melanoma, nevus, seborrheic keratosis), each paired with an expert-annotated binary lesion mask. The redundancy, aggregation, and statistical analyses are computed on the full 600-image test split; the meta-evaluation runs on

a fixed 64-image sample (Section 4.3). A fixed, reproducible 12-image pilot subset of the test split was used to validate the pipeline plumbing prior to the full run; Table 4.2 contrasts the two scales, and the axiom-verification check (Table 4.6) is also reported at that pilot scale.

#### 4.1.2. Models

Two classifiers act as the predictors whose decisions are explained: ResNet-18 (final `fc`  $\rightarrow$  `Linear(512, 3)`) and SqueezeNet 1.1 (final classifier  $\rightarrow$  `Conv2d(512, 3, 1, 1)`). Both are trained with the protocol of Section 3.3 (25 epochs, SGD with momentum, class-weighted cross-entropy, StepLR, fixed seed 42). Validation accuracy and trained-weight checksums are reported in Table 4.1.

**Tab. 4.1.** Predictive models under evaluation and their training outcome. Best validation accuracy is taken from the 25-epoch training logs; test-set accuracy is marked *n/a* as the 600-image test split was used for explanation evaluation (with the model’s own prediction as the attribution target), not for separate classification scoring.

| Model          | Head                 | Best val. accuracy | Test accuracy |
|----------------|----------------------|--------------------|---------------|
| ResNet-18      | Linear(512, 3)       | 75.3%              | <i>n/a</i>    |
| SqueezeNet 1.1 | Conv2d(512, 3, 1, 1) | 74.0%              | <i>n/a</i>    |

#### 4.1.3. Attribution Methods

Seven FAE methods are evaluated (Section 3.4): Saliency, Integrated Gradients, Grad-CAM, Guided Backpropagation, DeepLift, LRP (gradient-based), and Occlusion (perturbation-based). LIME and KernelSHAP were excluded from the experiment for runtime reasons, as discussed in Section 3.4.3.

#### 4.1.4. Metrics, Hardware, and Reproducibility

The twelve metrics of Table 3.1 are computed with the hyperparameters listed there. Each experiment is driven by a YAML configuration under `configs/` with fixed random seeds, so that the redundancy matrix, the meta-evaluation scores, and the aggregated rankings are reproducible from the raw per-(model, image, FAE, metric) score table (`results/full_run_7fae_12metrics_600.csv`; the pilot counterpart is `results/vertical_slice_7fae_12metrics.csv`).

**Tab. 4.2.** Experiment scale: pilot vs. full run.

| Quantity    | Pilot | Full run |
|-------------|-------|----------|
| Test images | 12    | 600      |
| Models      | 2     | 2        |
| FAE methods | 7     | 7        |
| Metrics     | 12    | 12       |
| Score rows  | 2,016 | 100,800  |

## 4.2. Metric Redundancy Analysis (Contribution 1)

This section addresses whether the twelve metrics carry independent signal or duplicate one another. Following Decision D4, for each model the metrics are first variance-pre-screened (a metric is dropped if, within a model group, the non-NaN fraction falls below 0.20 or the standard deviation falls below  $10^{-6}$ ), then the Spearman rank-correlation matrix is computed across the test images, and pairs with  $|\rho| > 0.85$  are greedily pruned, retaining in each pair the metric less correlated with the remainder. The surviving set is the reduced, non-redundant subset  $M^*$  that feeds the aggregation stage.

**4.2.0.0.1 Pre-screen.** Two metrics are removed before correlation analysis on structural grounds: *Completeness* (constant at 0.0 for the methods that satisfy the axiom, hence zero-variance and uninformative for ranking) and *Non-Sensitivity* (disabled by default due to  $O(d)$  forward-pass cost); *MPRT* was disabled for the full run outright (Section 3.5). The remaining nine metrics enter the matrix. The exclusion of *Completeness* is justified by the axiom-verification result (Table 4.6) rather than by correlation.

**4.2.0.0.2 Correlation matrices.** Tables 4.3 and 4.4 hold the within-category redundancy summaries for ResNet-18 and SqueezeNet, derived from `results/redundancy_matrix_resnet18.csv` and `results/redundancy_matrix_squeezenet.csv`. The full  $9 \times 9$  Spearman matrices are reproduced in those CSVs; only the summary and the pruned pairs are shown here for legibility.

**Tab. 4.3.** Redundancy summary — ResNet-18: within-category mean  $|\rho|$  at the  $|\rho| > 0.85$  pruning threshold (full run, 600 images).

| Category      | $n$ metrics | Mean $ \rho $ |
|---------------|-------------|---------------|
| Complexity    | 2           | 0.999         |
| Robustness    | 2           | 0.934         |
| Localization  | 2           | 0.248         |
| Faithfulness  | 2           | 0.052         |
| Randomization | 1           | <i>n/a</i>    |

**Tab. 4.4.** Redundancy summary — SqueezeNet 1.1: within-category mean  $|\rho|$  at the  $|\rho| > 0.85$  pruning threshold (full run, 600 images).

| Category      | $n$ metrics | Mean $ \rho $ |
|---------------|-------------|---------------|
| Complexity    | 2           | 0.993         |
| Robustness    | 2           | 0.976         |
| Localization  | 2           | 0.208         |
| Faithfulness  | 2           | 0.022         |
| Randomization | 1           | <i>n/a</i>    |

**4.2.0.0.3 Pruning outcome and  $M^*$ .** On the full 600-image run the Complexity pair (complexity / sparseness,  $|\rho| \approx 0.99$  on both models) and the Robustness pair (avg-/max-sensitivity,  $|\rho| \approx 0.93$  on ResNet-18 and 0.98 on SqueezeNet) are strongly redundant, whereas the Faithfulness and Localization pairs are nearly orthogonal ( $|\rho| < 0.25$  in every case). The greedy  $|\rho| > 0.85$  pruning therefore drops one metric from each redundant pair in both models (complexity in favour of sparseness; avg-sensitivity in favour of max-sensitivity). The resulting reduced set  $M^*$  is *identical* for the two models and comprises the same seven metrics: *faithfulness\_correlation*, *max\_sensitivity*, *pixel\_flipping*, *pointing\_game*, *random\_logit*, *relevance\_mass\_accuracy*, and *sparseness* (Tables 4.3–4.4). That the two architectures yield the same  $M^*$  is the principal evidence for Contribution 1: the redundancy structure of these metrics is a property of the metric battery rather than of the classifier. The pruning also costs no discriminative signal in practice: the per-method mean-effectiveness ranking computed over the full nine-metric battery and over  $M^*$  (uniform weights) are identical on both models (Spearman  $\rho = 1.00$ ). Table 4.5 records the set carried forward into aggregation.

**Tab. 4.5.** Reduced metric set  $M^*$  after redundancy pruning (full run, 600 images).  
The set is identical for both models.

| Set              | Metrics retained                                                      |
|------------------|-----------------------------------------------------------------------|
| ResNet-18 (7)    | f_corr, max_sens, pixel_flip, point_game, rand_logit, RMA, sparseness |
| SqueezeNet (7)   | f_corr, max_sens, pixel_flip, point_game, rand_logit, RMA, sparseness |
| Intersection (7) | f_corr, max_sens, pixel_flip, point_game, rand_logit, RMA, sparseness |

**Tab. 4.6.** Axiom verification (Completeness). All methods that satisfy the axiom by construction pass exactly. Pilot,  $n = 12$  per cell.

| Model                  | FAE method           | Mean violation | Passes |
|------------------------|----------------------|----------------|--------|
| ResNet-18 / SqueezeNet | Integrated Gradients | 0.000          | Yes    |
| ResNet-18 / SqueezeNet | DeepLift             | 0.000          | Yes    |
| ResNet-18 / SqueezeNet | LRP                  | 0.000          | Yes    |

### 4.3. Meta-Evaluation: Metric Reliability (Contribution 2)

The redundancy analysis identifies which metrics overlap; meta-evaluation identifies which metrics can be *trusted*. Each metric in  $M^*$  is subjected to this thesis’s estimators of the two MetaQuantus reliability properties (Section 3.7.2): Noise Resilience (NR), measuring score stability under semantically meaningless input/weight perturbation, and Adversarial Reactivity (AR), measuring whether the score reacts systematically as the explanation is deliberately degraded. The combined reliability  $r_k = \frac{1}{2}(\text{NR}_k + \text{AR}_k)$  is the discount coefficient used in Equation (3.8).

Unlike every other analysis in this chapter, the meta-evaluation runs on a fixed, seed-determined **64-image random sample** of the test split per (model, FAE) cell (5 noise seeds, 5 degradation levels); the extended-AR completion for *max\_sensitivity* and *random\_logit* used a fixed **12-image split-order sample** distinct from the 64-image draw, so those two metrics’ combined reliability mixes the two samples. The reliability estimates are correspondingly coarse, and are treated as such: values are reported to two decimals, the between-cell dispersion is reported alongside each mean, and the robustness of the downstream MQ-discount finding to these estimates is checked below.

Table 4.7 reports NR, AR, and combined reliability per metric, averaged over the (model, FAE) cells in which each metric is defined, read from `results/meta_evaluation_reliability.csv`. The runtime column documents the meta-evaluation cost per cell at these sample sizes (the full-run evaluation cost per metric is substantially larger and scales with the 600-image split).

**Tab. 4.7.** Metric reliability (NR, AR, combined  $r_k$ ) estimated with this thesis’s MetaQuantus-inspired protocol, averaged over the 14 (model, FAE) cells; SD is the between-cell standard deviation of  $r_k$ . NR and AR were computed on a fixed 64-image sample of the test split (5 noise seeds, 5 degradation levels); the extended-AR completion for *max\_sensitivity* and *random\_logit* (degradation injected into the explanation function, which those metrics re-invoke internally; Section 4.3) used a 12-image sample. Values are reported to two decimals to reflect the estimator’s granularity. Runtime is the meta-evaluation cost per (model, FAE) cell at these sample sizes, not the full-run evaluation cost.

| Metric                   | NR   | AR   | Combined $r_k$ | SD( $r_k$ ) | Runtime (s) |
|--------------------------|------|------|----------------|-------------|-------------|
| faithfulness_correlation | 0.56 | 0.60 | 0.58           | 0.28        | 151.5       |
| max_sensitivity          | 1.00 | 0.94 | 0.97           | 0.09        | 128.2       |
| pixel_flipping           | 0.99 | 0.77 | 0.88           | 0.15        | 186.7       |
| pointing_game            | 0.96 | 0.66 | 0.81           | 0.15        | 40.7        |
| random_logit             | 1.00 | 0.45 | 0.72           | 0.15        | 127.2       |
| relevance_mass_accuracy  | 0.59 | 0.51 | 0.55           | 0.22        | 40.5        |
| sparseness               | 0.99 | 0.94 | 0.97           | 0.09        | 40.4        |

The reliability table is read as follows: a metric with high NR but low AR is stable but insensitive (it does not react when the explanation degrades); a metric with low NR is noisy. Among the  $M^*$  metrics, *max\_sensitivity* ( $r_k \approx 0.97$ ), *sparseness* (0.97), and *pixel\_flipping* (0.88) are the most reliable, combining high noise resilience with strong adversarial reactivity; *pointing\_game* (0.81) and *random\_logit* (0.72) follow. *Faithfulness\_correlation* (0.58) and *relevance\_mass\_accuracy* (0.55) are the least reliable: both are noise-resilient only moderately (NR  $\approx 0.56$ – $0.59$ ) and react comparatively weakly to deliberate degradation (AR  $\approx 0.51$ – $0.60$ ). The dispersion column tempers these point estimates: the two least reliable metrics are also the least consistently estimated (SD( $r_k$ ) = 0.28 for *faithfulness\_correlation*, whose per-cell values span 0.05–0.98), whereas the most reliable metrics are also the most stable across cells (SD  $\approx 0.09$ ).

Two audits qualify the AR leg. First, because AR is the *absolute* Spearman correlation, a metric reacting in the unexpected direction would still earn credit; auditing the signed monotonicity of the 42 extended-AR cells finds 4 such cells (three for *random\_logit*, one for *max\_sensitivity*, with  $|\rho_S| \leq 0.7$ ) — their contribution is disclosed rather than re-scored, and a direction-rectified AR is noted as future work. Second, a leave-one-cell-out jackknife of the per-metric  $\lambda_k$  (recomputing the MQ-discounted ResNet-18 ranking with each of the seven FAE cells removed from the reliability average) preserves the headline reordering in all seven replicates: LRP remains second and the top-2 order is unchanged, so the MQ-discount finding of Section 4.4 is not an artefact of any single reliability cell. A harsher perturbation—shifting the two extension-measured AR values by  $\pm 0.2$ , twice the estimator’s granularity—preserves the reordering but can move LRP between second and third place; the robust statement is therefore that

the discount lifts LRP from fourth into the top three while Occlusion stays first, with the exact position contingent on the coarse AR estimate.

**4.3.0.0.1 Extending AR to explanation-recomputing metrics.** The MetaQuantus AR protocol degrades the explanation by zeroing a growing fraction of the attribution array. Two  $M^*$  metrics ignore that array: *max\_sensitivity* and *random\_logit* re-compute attributions internally through `explain_func`, so the array degradation never reaches them and their AR was initially unmeasurable (an earlier run of this pipeline reported *n/a* for both). The AR test was therefore extended: for such metrics the degradation is additionally injected into the explanation function itself—each invocation zeroes a fresh random fraction of the explainer’s output, so the metric’s internal re-explanations lose signal at the same rate as the explicit attribution (degradation capped at 0.9, since a fully zeroed attribution is rejected by the metric implementations). With this extension, every computable metric carries a combined reliability (126 of 168 reliability rows are valid; the remaining 42 belong to the three structurally excluded metrics—MPRT, Non-Sensitivity, and the zero-variance Completeness). The completed discount is no longer a near-no-op: it shifts the effectiveness magnitudes by mean  $|\Delta| \approx 0.07$  relative to Autoweighted and reorders the ResNet-18 mid-field ( $\rho(\text{AW}, \text{MQ}) = 0.79$ ; LRP rises from fourth to second at the point estimate) while leaving SqueezeNet nearly unchanged ( $\rho = 0.96$ , a single adjacent Saliency  $\leftrightarrow$  Occlusion swap; Table 4.10).

## 4.4. Aggregation and Ranking Comparison (Contributions 2–3)

This section measures whether *how* metrics are combined changes the conclusion about which FAE method is best. Three weighting schemes are applied to the same normalised scores over  $M^*$ : (i) **uniform** ( $w_k = 1/|M^*|$ ); (ii) **Autoweighted** (inverse-coefficient-of-variation weights, Equation (3.6)); (iii) **MQ-discounted** (Equation (3.8), the Autoweighted weights attenuated by reliability). A single-metric baseline (Faithfulness Correlation alone, `effectiveness_single_fc`) is included to quantify how far a naive one-metric ranking departs from the multi-criteria index. All four columns are present in `results/ranking_comparison.csv`. As noted in Section 4.3, the MQ-discount column applies a genuine reliability reweighting (mean  $|\Delta| \approx 0.07$  relative to Autoweighted): it reorders the ResNet-18 mid-field ( $\rho = 0.79$ ) while leaving SqueezeNet nearly unchanged ( $\rho = 0.96$ ).

**Tab. 4.8.** Mean effectiveness index  $E(\Phi)$  per FAE method under each weighting scheme (ResNet-18, averaged over test images).

| FAE method           | Uniform | Autoweighted | MQ-discount | Single-FC |
|----------------------|---------|--------------|-------------|-----------|
| integrated_gradients | 0.002   | 0.035        | 0.119       | -0.013    |
| saliency             | 0.062   | 0.062        | 0.130       | 0.334     |
| gradcam              | 0.133   | 0.023        | 0.082       | 0.812     |
| guided_backprop      | -0.035  | -0.060       | 0.054       | -0.002    |
| deep_lift            | -0.063  | -0.006       | 0.079       | -0.032    |
| lrp                  | -0.130  | 0.007        | 0.141       | -0.013    |
| occlusion            | 0.164   | 0.123        | 0.178       | 0.658     |

**Tab. 4.9.** Mean effectiveness index  $E(\Phi)$  per FAE method under each weighting scheme (SqueezeNet, averaged over test images).

| FAE method           | Uniform | Autoweighted | MQ-discount | Single-FC |
|----------------------|---------|--------------|-------------|-----------|
| integrated_gradients | -0.189  | -0.033       | 0.018       | -0.025    |
| saliency             | -0.001  | 0.058        | 0.102       | 0.718     |
| gradcam              | 0.142   | 0.124        | 0.162       | 1.435     |
| guided_backprop      | -0.127  | -0.004       | 0.060       | 0.096     |
| deep_lift            | -0.205  | -0.058       | -0.005      | 0.019     |
| lrp                  | -0.189  | -0.037       | 0.015       | -0.083    |
| occlusion            | 0.086   | 0.066        | 0.084       | 1.166     |

**Tab. 4.10.** Induced FAE-method ranks under each weighting scheme, and rank-stability summary (Spearman rank correlation between per-method mean- $E$  rankings; rank inversions summed over both models).

| Comparison                       | Uniform vs. AW | Uniform vs. MQ | AW vs. MQ | Single-FC vs. MQ |
|----------------------------------|----------------|----------------|-----------|------------------|
| Spearman rank corr. (ResNet-18)  | 0.750          | 0.321          | 0.786     | 0.286            |
| Spearman rank corr. (SqueezeNet) | 0.964          | 0.929          | 0.964     | 0.857            |
| # rank inversions                | 6              | 10             | 4         | 11               |

**4.4.0.0.1 The ranking shifts with the weighting scheme.** The headline finding of Contributions 2–3 is that *the induced FAE ranking depends measurably on how the metrics are combined—although the top-ranked method proves robust to that choice*. On ResNet-18 (Table 4.8) Occlusion leads under all three multi-metric schemes ( $E = 0.164$  uniform,  $0.123$  Autoweighted,  $0.178$  MQ-discount), but the mid-field

reorders substantially: Grad-CAM drops from second under uniform weights ( $E = 0.133$ ) to fourth under Autoweighted (0.023) and fifth under the MQ-discount (0.082), while under the MQ-discount LRP rises to second (0.141). The Spearman rank correlation between the uniform and Autoweighted per-method orderings is  $\rho = 0.750$  for ResNet-18, which contributes five of the six discordant method-pairs that Table 4.10 reports summed over both models. SqueezeNet is more stable (Table 4.9)—Grad-CAM is first and Occlusion second under both schemes ( $E = 0.142$  and  $0.086$  uniform,  $0.124$  and  $0.066$  Autoweighted) and the orderings agree closely ( $\rho = 0.964$ ). The single-metric baseline (Faithfulness Correlation alone) diverges most sharply of all on ResNet-18 (Single-FC vs. MQ-discount  $\rho = 0.286$  on ResNet-18,  $0.857$  on SqueezeNet, with eleven discordant pairs summed over both models), which confirms that a one-metric ranking is not a safe proxy for the multi-criteria index. Two further observations are notable. First, the two methods that lead on both models under uniform weighting—Occlusion and Grad-CAM—are precisely the two coarse, low-resolution attribution methods; where the weighting scheme changes the picture at all, it is Grad-CAM whose position is most sensitive (second under uniform to fourth under Autoweighted and fifth under the MQ-discount on ResNet-18). Second, the divergence is architecture-dependent: the reordering is pronounced on ResNet-18 and mild on SqueezeNet. This sensitivity to the weighting choice is precisely the phenomenon the framework is designed to surface; its interpretation is carried into Chapter 5.

## 4.5. Individual vs. Ensembled Attributions (Contribution 3)

Beyond comparing individual methods, the framework is applied to attributions produced by NormEnsembleXAI (Decision D6: Mean aggregation with Second Moment normalisation). The hypothesis under test is that an ensembled attribution achieves a higher—or at least more balanced—effectiveness index than its constituent individual methods, particularly on the Robustness axis where single methods are known to be unstable.

**4.5.0.0.1 Comparison protocol.** Each side’s effectiveness is a uniform NaN-aware mean over the seven  $M^*$  metrics, per (model, image), after Second-Moment scaling with a *pooled* RMS per (model, metric)—the individual-method scores and the ensemble scores share one scale, so a level difference between the two populations is visible. (An earlier per-source variant, in which each population was scaled by its own RMS, is blind to uniform level differences because  $\text{mean}(x)/\text{RMS}(x) = 1/\sqrt{1 + \text{CV}^2}$  for a positive-valued metric; it is retained only as a robustness variant below.) Four baselines are reported: the *mean* individual method; the *best fixed* method (the single strongest method per model); the best fixed *gradient* member (the strongest method actually inside the ensemble); and a per-image *oracle* that picks the best of all seven methods separately for every image—the strongest conceivable baseline, which may switch methods image-by-image and includes Occlusion, a non-member.

**Tab. 4.11.** Effectiveness index  $E(\Phi)$  of the NormEnsembleXAI-ensembled attribution against four individual-method baselines, per model (mean over  $n = 600$  test images per model; uniform weights over the seven  $M^*$  metrics; POOLED Second-Moment normalization, i.e. ensemble and individual scores share one RMS per (model, metric), so level differences are visible). The ensemble aggregates the six gradient-based methods; “oracle” is the per-image best over all seven methods (it may switch methods per image and includes Occlusion, which is not an ensemble member); “best fixed” is the single strongest method per model, and “best gradient” the strongest actual ensemble member.

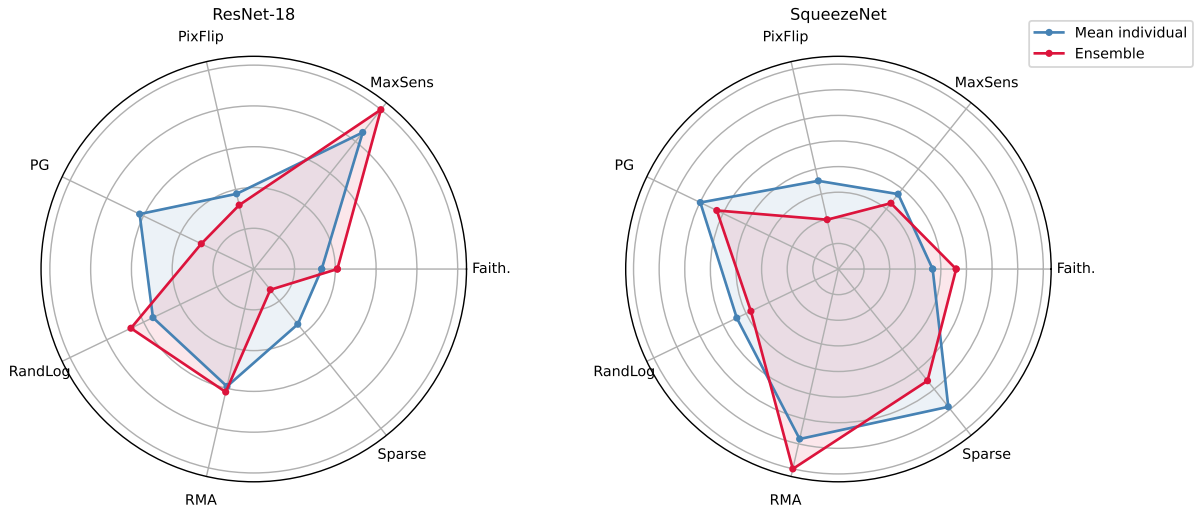
| Model      | Ensembled | Mean ind. | Best fixed        | Best gradient   | Oracle |
|------------|-----------|-----------|-------------------|-----------------|--------|
| ResNet-18  | -0.012    | 0.022     | 0.169 (occlusion) | 0.137 (gradcam) | 0.299  |
| SqueezeNet | -0.066    | -0.067    | 0.146 (gradcam)   | 0.146 (gradcam) | 0.213  |

**4.5.0.0.2 The ensemble does not beat any baseline.** Over all seven  $M^*$  metrics (Robustness included) and the full 600 images, Table 4.11 is unambiguous. The ensembled attribution is significantly *below* the mean individual method on ResNet-18 ( $-0.012$  vs.  $0.022$ ) and statistically indistinguishable from it on SqueezeNet ( $-0.066$  vs.  $-0.067$ ;  $p = 0.74$ , Table 4.13); it is decisively below the best fixed method on both models ( $0.169$  for Occlusion on ResNet-18,  $0.146$  for Grad-CAM on SqueezeNet), below the best gradient member it actually contains, and far below the per-image oracle ( $0.299 / 0.213$ ). Averaging attribution maps does not, on this data, buy effectiveness: whatever stability the mean map gains is offset by losses on other axes, and the averaged map is not more effective than its strongest constituent.

**4.5.0.0.3 Normalization-conditionality of the earlier verdict.** Under the per-source variant the SqueezeNet comparison *appears* favourable to the ensemble ( $0.075$  vs.  $-0.069$  against the mean individual method, nominally significant). That apparent win does not survive placing both populations on a common scale, and is therefore reported as an artefact of the normalization choice rather than as a finding—a concrete instance of the normaliser-conditionality caveat of Chapter 5. The mechanism is traceable: the individual methods’ raw relevance-mass-accuracy scores are heavy-tailed (Section 5.4 of Chapter 5 quantifies the out-of-range values), so their per-source RMS is inflated and their normalized RMA crushed relative to the well-behaved ensemble scores. Under a winsorized pooled scale—a third normalizer that clips each (model, metric) pool at its 1st/99th percentile before scaling (results/individual\_vs\_ensemble\_winsorized.csv)—the SqueezeNet comparison turns significantly *against* the ensemble ( $-0.038$  vs.  $-0.017$ ,  $W = 67206$ ,  $p < 10^{-3}$ ), so the “beats none of the baselines” verdict holds under all three normalizers while the direction of the mean-comparison on SqueezeNet is normalizer-dependent. The practical reading is that NormEnsembleXAI’s Mean ensemble is not an improvement on this task under any reported baseline, and never a substitute for identifying the per-model best method.

**4.5.0.0.4 Caveat: ensemble scope.** The ensemble is formed over the six gradient-based methods only; Occlusion was excluded as an ensemble member because ensembling a perturbation method inside the Robustness metrics—which re-attribute the ensemble many times per image—is prohibitively costly. The comparison otherwise uses the full 600 test images and all seven  $M^*$  metrics, so it is a full-scale verdict modulo that single excluded member.

Per-axis effectiveness (one effectiveness value per metric category for the ensembled vs. individual attributions) is reported in Figure 4.1 so that trade-offs—e.g. a gain in Robustness purchased at the cost of Localization—are visible rather than hidden by the scalar index.



**Fig. 4.1.** Per-metric comparison of the NormEnsembleXAI ensemble (crimson) against the mean individual attribution method (blue) over the seven  $M^*$  metrics on the full 600 images, normalised per metric against the method range and rectified by metric direction, so a larger radius is always better. On both models the ensemble trades gains on some metrics for losses on others rather than dominating uniformly—which is why its scalar effectiveness never exceeds the individual baselines of Table 4.11.

## 4.6. Statistical Significance (Contribution 4)

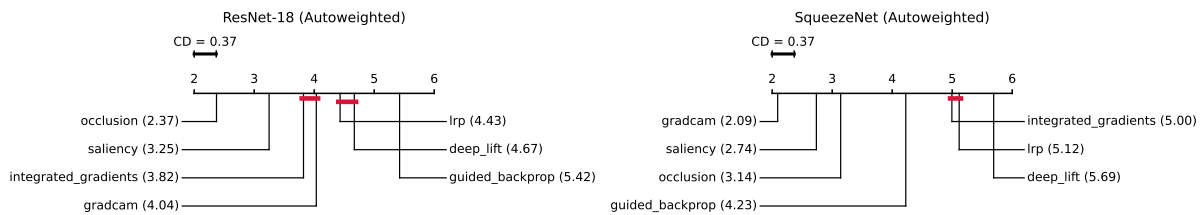
The final stage tests whether the observed ranking differences are statistically significant rather than artefacts of sampling (Decision D5). Two complementary procedures are applied to the per-image effectiveness distributions:

- **Friedman test + Nemenyi post-hoc** across all FAE methods. The Friedman test detects whether at least one method differs in mean rank across the test images; the Nemenyi post-hoc identifies which pairs differ, visualised as a critical-difference (CD) diagram.
- **Wilcoxon signed-rank test** for the paired comparison of individual vs. NormEnsembleXAI-ensembled attributions (Section 4.5), which is inherently paired per image.

**Tab. 4.12.** Friedman omnibus test over the seven FAE methods, per model and weighting scheme (blocks are the 600 test images).  $W$  is Kendall’s coefficient of concordance,  $\chi^2/(N(k-1))$ , the standardized effect size.

| Model      | Weighting    | Friedman $\chi^2$ | $p$ -value  | Kendall’s $W$ | Sig. ( $\alpha=0.05$ ) |
|------------|--------------|-------------------|-------------|---------------|------------------------|
| ResNet-18  | Uniform      | 830.24            | $< 10^{-3}$ | 0.23          | Yes                    |
| ResNet-18  | Autoweighted | 759.65            | $< 10^{-3}$ | 0.21          | Yes                    |
| ResNet-18  | MQ-discount  | 469.20            | $< 10^{-3}$ | 0.13          | Yes                    |
| SqueezeNet | Uniform      | 1571.34           | $< 10^{-3}$ | 0.44          | Yes                    |
| SqueezeNet | Autoweighted | 1430.30           | $< 10^{-3}$ | 0.40          | Yes                    |
| SqueezeNet | MQ-discount  | 1217.74           | $< 10^{-3}$ | 0.34          | Yes                    |

The Friedman omnibus test rejects the null hypothesis of equal method ranks decisively on both models (Table 4.12): ResNet-18 yields  $\chi^2 = 759.6$  and SqueezeNet  $\chi^2 = 1430.3$ , both with  $p < 10^{-3}$  over the 600 image blocks under the Autoweighted scheme (uniform:  $\chi^2 = 830.2 / 1571.3$ ; MQ-discount:  $469.2 / 1217.7$ ; likewise  $p < 10^{-3}$ ). The standardized effect sizes put these in perspective: Kendall’s  $W$  indicates small-to-moderate concordance of the per-image rankings on ResNet-18 ( $W = 0.13$ – $0.23$  across schemes) and moderate concordance on SqueezeNet ( $0.34$ – $0.44$ )—the method differences are unambiguous but far from deterministic at the level of individual images. Twelve omnibus and paired tests are run in this chapter; no formal multiplicity correction is applied, which is immaterial at the observed  $p < 10^{-10}$  but is noted for completeness (the one non-significant result below is reported as such, not screened out). At least one FAE method therefore differs significantly in mean rank, so the Nemenyi post-hoc and CD diagram (Figure 4.2) are warranted; the per-method mean ranks feeding that diagram are in `results/cd_summary.csv`.



**Fig. 4.2.** Nemenyi critical-difference diagram over the seven FAE methods under the Autoweighted scheme, per model (mean-rank axis; methods joined by a crimson bar are not significantly different at  $\alpha = 0.05$ ). Occlusion leads on ResNet-18 and Grad-CAM on SqueezeNet; the weighting-dependent mid-field re-ordering of ResNet-18 (Grad-CAM second under uniform weights, fourth here) is visible against the more stable SqueezeNet order.

**Tab. 4.13.** Wilcoxon signed-rank tests (paired per image,  $n = 600$  per model, pooled normalization): the NormEnsembleXAI-ensembled attribution against the mean individual method, the best fixed method, and the per-image oracle.  $r$  is the matched-pairs rank-biserial correlation (positive = ensemble higher).

| Model      | Pair                  | $W$     | $p$ -value  | $r$   | Significant |
|------------|-----------------------|---------|-------------|-------|-------------|
| ResNet-18  | vs. mean individual   | 62587.0 | $< 10^{-3}$ | -0.31 | Yes         |
| ResNet-18  | vs. best fixed method | 27972.0 | $< 10^{-3}$ | -0.69 | Yes         |
| ResNet-18  | vs. per-image oracle  | 562.0   | $< 10^{-3}$ | -0.99 | Yes         |
| SqueezeNet | vs. mean individual   | 88718.0 | 0.736       | 0.02  | No          |
| SqueezeNet | vs. best fixed method | 12004.0 | $< 10^{-3}$ | -0.87 | Yes         |
| SqueezeNet | vs. per-image oracle  | 162.0   | $< 10^{-3}$ | -1.00 | Yes         |

The paired Wilcoxon tests (Table 4.13,  $n = 600$  per model, pooled normalization) quantify the ensemble result of Section 4.5. Against the *mean* individual method the ensemble is significantly *worse* on ResNet-18 ( $W = 62587$ ,  $p < 10^{-3}$ , rank-biserial  $r = -0.31$ ) and statistically indistinguishable on SqueezeNet ( $W = 88718$ ,  $p = 0.74$ ,  $r = 0.02$ ). Against the *best fixed* method it is decisively outperformed on both models ( $r = -0.69$  and  $-0.87$ ), and against the per-image oracle the losses are near-total ( $r = -0.99$  and  $-1.00$ ); all five of these comparisons are significant at  $p < 10^{-3}$  over the 600 paired images. The conclusion—the Mean ensemble beats none of the reported baselines, and does not reliably beat even the average individual method—is therefore statistically robust rather than a sampling artefact.

The broader interpretation of these tests is carried into the discussion of Chapter 5.



## 5. Discussion

This chapter interprets the experimental results of Chapter 4 in the light of the research questions, examines the threats to the validity of the conclusions, and delimits the scope within which the findings can be generalised. The discussion is organised so that the contribution of the proposed evaluation framework can be appraised *independently* of any particular ranking it produces: the framework is a methodological artefact whose value lies in how it structures the comparison, and that value does not stand or fall with the specific ordering of FAE methods that the full run reports.

### 5.1. Interpretation of the Rankings

This section interprets what the aggregated rankings of Section 4.4 imply for practitioners selecting an FAE method. The empirical yield is best read as a set of *cautionary* findings: the framework surfaces real instabilities in how FAE methods are ranked, and the two literature-derived enhancements it tests deliver a model-dependent rather than universal benefit. Each item below states the verdict the full 600-image run returns.

- **Does multi-criteria aggregation change the verdict?** Yes—but conditionally. Comparing the single-metric baseline (Faithfulness Correlation alone) against the multi-criteria index (Table 4.10) shows that a one-metric ranking is not a safe proxy for the aggregate: the Single-FC and MQ-discounted orderings correlate only weakly on ResNet-18 ( $\rho = 0.29$ ) and imperfectly on SqueezeNet ( $\rho = 0.86$ ). A method that leads on Faithfulness alone is not guaranteed to survive once Robustness, Localization, and Randomization enter the aggregate. This vindicates the multi-criteria premise of the thesis: the scalar leaderboard one reads off a single faithfulness metric can differ materially from the one a balanced index returns.
- **Effect of the weighting scheme.** The headline—and the most important cautionary finding—is that *how much the FAE ranking changes with the weighting scheme is itself model-dependent*. On ResNet-18 the uniform and Autoweighted method-rankings agree only moderately (Spearman  $\rho = 0.75$ , five discordant method-pairs): Occlusion stays first under both schemes, but Grad-CAM drops from second under uniform weights to fourth under Autoweighted while Saliency and Integrated Gradients move up. On SqueezeNet the same two schemes agree almost perfectly ( $\rho = 0.96$ ), with Grad-CAM first under both. The weighting choice can therefore reorder much of

the field on one architecture while leaving the other nearly untouched. The most plausible mechanism is the inverse-variance character of the Autoweighted scheme: its weights are driven by each model’s per-metric score-variance structure, so two models with different variance profiles redistribute weight across  $M^*$  differently, and only where that redistribution is large does the ranking move. This is evidence that aggregation *can* reorder rankings materially, not that it reliably does—a warning against trusting any single-number FAE leaderboard, but not a licence to assume the weighting always matters.

- **Does the MetaQuantus reliability discount change anything?** Yes—moderately, and model-dependently, once the meta-validation is complete. With all seven  $M^*$  metrics carrying a measured combined reliability, the MQ-discounted ordering departs from Autoweighted on ResNet-18 ( $\rho = 0.79$ : LRP rises from fourth into the top three (second at the point estimate) and Grad-CAM falls to fifth) while SqueezeNet moves by a single adjacent swap ( $\rho = 0.96$ ); the mean effectiveness shift is  $|\Delta| \approx 0.07$ . Mechanically, the discount redistributes weight toward the high-reliability metrics (max-sensitivity  $r_k \approx 0.97$ , sparseness 0.97, pixel-flipping 0.88) and away from the low-reliability ones (random-logit 0.72, faithfulness-correlation 0.58, relevance-mass accuracy 0.55). The discount is therefore a consequential modelling choice on one architecture and nearly inert on the other—the same conditional pattern as the uniform-vs-Autoweighted comparison (Section 5.2).
- **Per-category trade-offs.** The ensembling comparison (Section 4.5) shows that NormEnsembleXAI does not buy any improvement on this task: on both models the ensemble trades gains on some metric categories for losses on others rather than dominating, and once ensemble and individual scores are placed on a common scale its scalar effectiveness beats none of the reported baselines—significantly worse than the mean individual method on ResNet-18, indistinguishable from it on SqueezeNet, and decisively below the best fixed method on both. A nominal SqueezeNet advantage under the earlier per-source normalization proved to be an artefact of that normalization choice, which is exactly why the per-category profiles and the normalization protocol are reported alongside the scalar.
- **Cross-model consistency.** The two architectures agree on the one result that matters most for Contribution 1—the reduced metric set  $M^*$  is *identical* across ResNet-18 and SqueezeNet—but disagree on almost everything downstream: the ranking divergence is large on ResNet-18 and negligible on SqueezeNet, and the ensemble is significantly worse than the average method on ResNet-18 yet indistinguishable from it on SqueezeNet. Persistent agreement on  $M^*$  strengthens the claim that the redundancy structure is a metric-battery property; persistent disagreement elsewhere is itself the finding, and a caution against treating any single model’s ranking as the framework’s verdict.

## 5.2. Implications of the Redundancy and Reliability Findings

The redundancy and reliability analyses are complementary filters acting on the same metric set, and their interaction is the point of methodological interest. On the full 600-image run the two filters give different yields: the redundancy filter is the strongest and cleanest result of the thesis, while the reliability filter—once turned into the MQ-discount—produces a moderate, architecture-dependent reweighting of the ranking.

- **Redundant categories.** The full-run correlation matrix shows two metric pairs collapsing to a single effective dimension on *both* architectures: Complexity (complexity  $\leftrightarrow$  sparseness,  $|\rho| = 0.999$  on ResNet-18 and 0.993 on SqueezeNet) and Robustness (avg-sensitivity  $\leftrightarrow$  max-sensitivity,  $|\rho| = 0.93$  and 0.98 respectively), whereas the Faithfulness and Localization pairs remain nearly orthogonal ( $|\rho| < 0.25$ ). The practical implication is concrete: reporting both members of a near-collinear pair overstates the evidence, because the two metrics constitute one vote, not two—an evaluator who quotes “twelve metrics” is in effect quoting roughly seven independent ones. That the greedy pruning yields an *identical* seven-metric  $M^*$  on two different architectures is the strongest single finding here: the redundancy is a property of the metric battery, not of the classifier, and it transfers. A practitioner can therefore halve the costly Robustness leg (avg-sensitivity pruned, at  $\sim 128$  s per (model, FAE) cell) and drop Complexity’s duplicate, with no loss of discriminative signal; the two expensive Faithfulness metrics are retained, as both carry independent signal.
- **Greedy cross-category pruning.** The current pruning is greedy and can in principle drop a metric because it correlates with a metric in a *different* category. On this data the consequence is benign—both pruned pairs are *within* a single category (Complexity; Robustness), so the greedy policy implemented as Decision D4 and a strictly within-category alternative coincide and yield the same  $M^*$ . The robustness of the result to that policy choice is reassuring, but it is a property of this dataset’s correlation structure and not a guarantee; a strictly within-category policy remains the more defensible default and is recommended for transfer to other domains.
- **Reliability vs. agreement, and what the MQ-discount does.** The MetaQuantus reliability ordering and the dispersion-based Autoweighted ordering are conceptually distinct, and the thesis proposed the MQ-discount precisely to let the former attenuate the latter. Empirically, the discount is consequential exactly where the two orderings disagree most: on ResNet-18 it shifts the per-method effectiveness by  $|\Delta| \approx 0.07$  on average and reorders the mid-field ( $\rho(\text{Autoweighted}, \text{MQ-discount}) = 0.79$ ; LRP fourth  $\rightarrow$  second), while on SqueezeNet it is nearly inert ( $\rho = 0.96$ ). Two caveats keep this claim honest. First, the AR leg for the two explanation-recomputing metrics required a protocol extension introduced by this thesis (degrading the explanation function itself, Section 4.3), so their reliability scores are measured under an extended, not the original, MetaQuantus protocol. Second, no ground truth says the MQ-discounted ranking

is *better*—the framework establishes that reliability weighting changes the verdict, not that the changed verdict is correct. The contribution is the measured answer to a question the literature had left assumed: a reliability-discounted aggregation *can* change the FAE ranking, and whether it does is architecture-dependent.

### 5.3. The Value of the Aggregation Framework Independent of the Rankings

A central claim of this thesis is that the proposed evaluation pipeline is a contribution in its own right, separable from whichever FAE method it happens to rank highest on ISIC 2017. Three arguments support reading the framework’s value independently of the specific rankings it produces.

First, the pipeline makes the configuration of an evaluation *explicit and reproducible*. The four-stage decomposition—redundancy-aware metric selection, meta-validation, direction-rectified normalisation, and weighted aggregation—separates the choices that an evaluator must make (which metrics, which normalisation, which aggregator, which weighting) from the method under test. Two researchers who disagree about the conclusion can now locate the disagreement in a named configuration decision rather than in an opaque, ad-hoc averaging of scores. This is precisely the property that Seth and Sankarapu [36] argue is required for an evaluation to function as an auditable, compliance-grade instrument, and it holds whatever the rankings turn out to be.

Second, the framework operationalises a defensible answer to the metric disagreement documented in the literature. Krishna et al. [10] and Tomsett et al. [15] show that XAI metrics frequently disagree and that the apparent rigour of reporting “many” metrics is illusory when those metrics are correlated. The redundancy stage answers this by collapsing correlated metrics to representatives; the meta-validation stage answers the companion concern—that some metrics are simply unreliable [14]—by discounting metrics that fail Noise Resilience or Adversarial Reactivity. The MQ-discounted Autoweighted operator fuses both filters into a single, data-adaptive weight derivation; whether that fusion measurably changes the ranking is an empirical question, and on this data the answer is architecture-dependent—material on ResNet-18, marginal on SqueezeNet (Section 5.2). The value of the framework here is that it let the question be *asked and answered* rather than assumed, and that the answer comes with its conditionality attached. The methodological claim—that aggregation should be preceded by redundancy pruning and that reliability weighting is worth testing—is itself testable and transferable whatever the leaderboard outcome on a particular dataset.

Third, the framework yields a *stable apparatus for asking conditional questions* rather than a single verdict. By reporting three aggregation operators (weighted mean, minimum, geometric mean) and three weighting schemes, the pipeline exposes how sensitive any conclusion is to the evaluator’s priorities. The minimum operator encodes a worst-dimension, safety-critical stance; the geometric mean penalises catastrophic failure on any axis; the weighted mean expresses a compensatory trade-off. A practitioner

in a high-stakes medical setting and a practitioner prototyping a research model can read different recommendations off the same computation. The framework’s contribution is to make that conditionality visible and principled, which is independent of the numerical outcome of any one configuration.

## 5.4. Threats to Validity

### 5.4.1. Internal Validity

Several design decisions could bias the conclusions drawn from within the study itself, independently of how far they generalise.

The most consequential internal threat concerns **metric failures and missing-value handling**. The Model Parameter Randomisation Test (MPRT) proved incompatible with this dataset under Quantus 0.6.0: during the pilot it yielded large numbers of NaN values—most acutely for SqueezeNet and for Grad-CAM on the randomised ResNet—and on ISIC 2017 its implementation ultimately failed on every (model, image) sample, so it was disabled outright for the full run (Section 3.5). The decision has since been vindicated by the metric’s own authors, who document methodological flaws in the original MPRT and propose repaired variants [42]; a future run should adopt Smooth- or Efficient-MPRT rather than the original test. When a metric is computable on only a biased subsample of images, any correlation or reliability estimate derived from it is suspect: an apparent architectural difference in, say, the MPRT–Sparseness relationship between the two models may be a genuine effect or merely an artefact of which images happened to produce a finite score. For this reason the Randomization category is represented in the aggregated index by RandomLogit alone. Similarly, Non-Sensitivity is excluded from the routine evaluation because its  $O(d)$  forward-pass cost forces aggressive input downsampling, which changes what the metric measures and makes its scores non-comparable with the per-pixel metrics; it is reported, if at all, only as a standalone diagnostic. Completeness is intrinsically defined only for the axiomatic methods (Integrated Gradients, DeepLift, LRP) and returns NaN elsewhere by construction; this is handled by per-metric masking rather than imputation, so that no method is credited or penalised on an inapplicable axis.

A second internal threat is the **quality of the predictors being explained**. The two classifiers are only modestly accurate: ResNet-18 reaches 75.3% best validation accuracy and SqueezeNet 74.0%, while ISIC 2017 is dominated by the nevus class, so the majority-class baseline is already around 65%—the predictors clear it by only a narrow margin. ResNet-18 moreover shows clear overfitting (98% train accuracy against 75.3% validation; SqueezeNet is better balanced at 78.5% train / 74.0% validation). The attributions therefore explain imperfect, partly over-fitted classifiers. This is a genuine validity consideration—an explanation of a flawed model inherits that model’s flaws—but it is not fatal to the thesis, because the predictive models are explicitly *not* the contribution: the unit of novelty is the evaluation pipeline, which is exercised identically whatever the classifier’s accuracy. The finding to carry forward is simply that the rankings reported here characterise these two specific, imperfect models, not an idealised classifier.

A third internal threat is the **target-class convention**. Attributions are computed for the model’s predicted class rather than the ground-truth label. This is the correct choice for a faithfulness study—the explanation should describe the decision the model actually made—but it interacts with misclassified images, where the explained logit corresponds to a wrong class and the Localization metrics are scored against a mask that need not align with what the model attended to. Given the modest accuracy above, a non-trivial fraction of test images are misclassified, which bounds the interpretability of the Localization results accordingly.

A fourth internal threat is the **extended AR protocol**, and it bears directly on the foundation of the MQ-discount. The Adversarial Reactivity of max-sensitivity and random-logit cannot be measured by degrading the attribution array (both metrics re-compute attributions internally and ignore it), so this thesis extends the MetaQuantus protocol: for those metrics the degradation is injected into the explanation function itself (Section 4.3). With the extension, all seven  $M^*$  metrics carry a measured combined reliability (126 of 168 reliability rows are valid; the invalid remainder belongs entirely to the three structurally excluded metrics). Three caveats accumulate here. First, the extension is this thesis’s own construction rather than part of the published MetaQuantus suite: the degraded-explainer AR measures reaction to a *stochastically corrupted explanation process*, which is analogous to, but not identical with, the original array-degradation test—and for the sensitivity metrics, whose very definition compares repeated explainer invocations, the independently drawn degradation masks mechanically induce part of the measured reactivity, so max-sensitivity’s top reliability rank should be read with particular caution. Second, the reliability estimates rest on small samples (64 images; 12 for the extension, drawn by a different rule than the 64) with a coarse-grained AR estimator, as disclosed in Section 4.3; the leave-one-cell-out jackknife reported there shows the ResNet-18 reordering is at least not an artefact of any single cell. Third, the direction-agnostic  $|\rho_S|$  awarded credit to four of the 42 audited cells that reacted in the unexpected direction. The reliability scores—and therefore the ResNet-18 reordering the MQ-discount produces—inheriting these caveats, and validating the extension (shared degradation masks per level, a direction-rectified AR, larger samples) is listed as future work.

A fifth internal threat is **post-selection inference**. The same 600 test images supply the correlation matrix that selects  $M^*$ , the normalisation statistics, the Autoweighted weights, and the per-image effectiveness values on which the Friedman and Wilcoxon tests are run. The significance tests therefore certify differences in a statistic whose configuration was itself tuned on the tested data, and their nominal  $p$ -values should be read in that light. Two mitigations apply: the selection and weighting steps do not optimise for any particular method ordering (they see only metric-level statistics, never the method labels being ranked), and the observed  $p$ -values ( $< 10^{-10}$ ) leave a wide margin. A held-out-split protocol—selecting  $M^*$  and the weights on one half of the test set and testing on the other—is the clean remedy and is listed as future work.

### 5.4.2. Construct Validity

Construct validity asks whether the chosen metrics actually capture the quality dimensions they are named for. The framework’s premise is that the six Quantus categories—Faithfulness, Robustness, Localization, Complexity, Randomization, and Axiomatic correctness—together span the relevant notion of explanation “effectiveness.” Two residual risks remain even after redundancy pruning and meta-validation.

The first is that meta-validation itself relies on proxies. Noise Resilience and Adversarial Reactivity [14] are plausible necessary conditions for a trustworthy metric, but they are not sufficient: a metric can be stable under noise and reactive to explanation degradation while still failing to track any human-meaningful sense of explanation quality. Independent meta-evaluations sharpen the point: using transparent models with ground-truth explanations, Miró-Nicolau, Capó, and Moyà-Alcover [43] find that Max- and Avg-Sensitivity fail to flag randomly generated explanations—a more critical verdict on the same Robustness metrics than the NR/AR tests reach here, and a reminder that a metric’s measured reliability depends on which meta-evaluation lens is applied. Because the entire study is functionally grounded—no clinician judgements enter the loop—there is an irreducible gap between “passes the meta-evaluation stress tests” and “measures what a domain expert would call a good explanation.” The framework narrows this gap but cannot close it without the human-grounded validation that is explicitly out of scope.

A related construct risk is that **metric implementations can emit out-of-range or heavy-tailed raw scores**. Relevance Mass Accuracy is nominally a mass fraction in  $[0, 1]$ , yet with signed attributions (`abs=False`) its denominator—the total signed relevance—can approach zero, and 26% of the valid full-run RMA values exceed 1 (maximum  $\approx 2 \times 10^3$ , concentrated in Integrated Gradients, DeepLift, Guided Backpropagation, and LRP). Max-Sensitivity similarly explodes where the re-attributed explanations are unstable (for LRP on ResNet-18 the median is  $\approx 52$  and the maximum  $\approx 1.9 \times 10^5$ ). Second Moment Scaling dampens but does not remove such heavy tails—an outlier-inflated RMS deflates every other score in its group—and this mechanism is precisely what produced the per-source ensemble artefact of Section 4.5. Robust or winsorized scaling, and an absolute-value RMA variant, are natural hardening steps for future work.

The second construct risk concerns **normalisation**. Second Moment Scaling was adopted from NormEnsembleXAI [11] because it aligns heterogeneous attribution scales without the outlier sensitivity of min–max scaling, but the choice is a modelling assumption about what makes two metric scores “comparable.” The ranking could in principle shift under  $z$ -score or min–max normalisation; any conclusion that is not stable across normalisers must be stated as normaliser-conditional. This is not a hypothetical caveat: the individual-vs-ensemble verdict on SqueezeNet changes from a nominally significant ensemble win to statistical indistinguishability when the per-source scaling is replaced by a pooled scale (Section 4.5)—the clearest demonstration in this thesis that a normalisation choice can manufacture or dissolve a finding. Conversely, the per-(model, metric) grouping used here has independent support: Edin et al. [44] show that raw faithfulness scores are not comparable across models at all unless normalised

per model, which is exactly the grouping this thesis applies before any cross-model statement is made. This caveat is sharpened by the nature of the effectiveness index itself.

The third construct risk is that **the effectiveness index is a relative, normaliser-conditional, unbounded quantity, not an absolute quality score**. Under the Autoweighted scheme it ranges roughly  $-0.49$  to  $0.61$  across per-image evaluations, and under the single-metric baseline it spans  $-2.65$  to  $4.02$ ; the sign and magnitude depend on the chosen normaliser and on the set of methods being compared. The index is therefore a *ranking device*—it orders methods within a fixed configuration—and must not be read as an absolute measure of explanation quality, nor compared across configurations as though it lived on a common scale. Every numerical effectiveness value reported in this thesis should be interpreted in that relative sense.

### 5.4.3. External Validity

External validity concerns the reach of the conclusions beyond the precise conditions of the experiment, and it is where the study is most deliberately bounded.

All evidence comes from a **single dataset and domain**: ISIC 2017 dermoscopy. The medical domain was chosen deliberately—it supplies the expert-annotated lesion masks that make the Localization category measurable at all, and it grounds the regulatory motivation of the thesis—but a single domain limits how far the redundancy and reliability *structure* can be assumed to transfer. The correlation between two faithfulness metrics, or the reliability of MPRT, may differ on natural images or on a different imaging modality; the framework is portable, but the particular  $M^*$  and weight vector it produces are not claimed to be.

The study uses **only two architectures**, ResNet-18 and SqueezeNet 1.1, both convolutional. Transformer-based classifiers route gradients differently and may interact differently with gradient-based attributions and with the layer-wise randomisation that MPRT performs; nothing in this study speaks to that regime. The two-model design does allow a minimal cross-architecture consistency check, and it earns its keep: the identical  $M^*$  across the two models is the thesis’s strongest transfer claim. But the same two-model design sharply limits the reach of the *divergence* findings. The headline result that aggregation can reorder the FAE ranking rests substantially on a single model—ResNet-18, where the uniform/Autoweighted correlation is  $\rho = 0.75$ , against  $\rho = 0.96$  on SqueezeNet—and the ensemble’s standing likewise differs between the two (significantly below the mean on ResNet-18, indistinguishable from it on SqueezeNet). A finding that is large on one of two architectures and negligible on the other cannot be generalised; it is a demonstration that the phenomenon *can* occur, not a measurement of how often it does. A wider panel of architectures is needed before the divergence is treated as anything more than possible.

Finally, the redundancy and reliability structure is estimated from a **single finite sample from one challenge distribution**. Even at the full 600 images, confidence intervals on the correlation and reliability estimates and a repeated-subsample stability analysis would be needed before the metric structure reported here is treated as a property of the domain rather than of this particular draw of images—and

the reliability estimates for the two explanation-recomputing metrics, obtained under the extended AR protocol, should be read with the internal-validity caveat above attached.

## 5.5. Limitations

Beyond the validity threats above, several scope limitations bound the contribution and are stated here so that the conclusions of Chapter 6 are read in context.

- **Restricted method panel.** The computational cost of fitting per-instance surrogate models (LIME) and sampling Shapley coalitions (KernelSHAP)—a cost the Robustness metrics multiply, since they re-invoke the explanation function for every perturbation sample—constrained the evaluated panel to the seven methods retained in Chapter 3 (Integrated Gradients, Saliency, Grad-CAM, DeepLift, Guided Backpropagation, LRP, and Occlusion). The two perturbation-based surrogate methods are therefore deferred rather than dismissed, and their integration is identified as future work.
- **Map-level explanations only.** The framework evaluates attribution *maps*. It does not address concept-level, prototype-based, or counterfactual explanations, whose quality criteria differ and would require a separate metric taxonomy.
- **Functionally grounded evaluation only.** Human-grounded validation—clinician studies of whether the highlighted regions are clinically sensible—is explicitly out of scope. The effectiveness index is a functional measure, and its alignment with expert judgement is an open question rather than an established result.
- **Aggregation to a scalar.** Collapsing a multi-criteria profile to a single effectiveness index is itself a modelling choice that can hide trade-offs between categories. The thesis mitigates this by reporting the underlying per-category profiles alongside the scalar index, but the scalar remains a summary, not a substitute for the profile.



## 6. Conclusions and Future Work

This thesis set out to evaluate Feature Attribution Explanation methods under multiple, potentially conflicting effectiveness criteria, treating the evaluation pipeline—rather than any new explanation method or predictive model—as the unit of contribution. The motivating problem is well documented: the proliferation of XAI evaluation metrics has not produced consensus, because the metrics disagree, partially duplicate one another, and are themselves of uncertain reliability. The response developed here is a single, reproducible pipeline that prunes redundant metrics, discounts unreliable ones, and aggregates the survivors into an effectiveness index with explicit, auditable configuration. This chapter summarises the outcome against each of the four contributions and outlines the directions that the work opens.

The empirical yield of the full 600-image run is, in candour, more cautionary than triumphant: the cleanest contribution is a redundancy result, while the two literature-derived enhancements the thesis set out to test—the reliability-discounted weighting and attribution ensembling—each deliver a model-dependent rather than universal benefit. These conditional findings are not a shortfall but the substance of the contribution—each warns against an assumption that practitioners and the prior literature make too readily.

### 6.1. Summary of Contributions

The four contributions map onto what the thesis set out to do as follows. Each paragraph states the methodological outcome—what was developed, proposed, or tested—and then the quantitative verdict the full run returns.

**6.1.0.0.1 Contribution 1 — Non-redundancy analysis of FAE evaluation metrics.** The thesis develops a correlation-based redundancy analysis across the six Quantus categories (Section 4.2), with a documented pruning rule, and derives a reduced, non-redundant metric set  $M^*$ . The methodological outcome is a transferable procedure for diagnosing and removing metric redundancy before any comparison is made, directly addressing the “many correlated metrics overstate evidence” problem identified by Tomsett et al. [15] and Pawlicki et al. [31]. *Verdict.* This is the strongest and cleanest result of the

thesis. On the full 600 images, two metric pairs proved near-perfectly redundant on both architectures—Complexity (complexity  $\leftrightarrow$  sparseness,  $|\rho| = 0.999$  on ResNet-18 /  $0.993$  on SqueezeNet) and Robustness (avg-  $\leftrightarrow$  max-sensitivity,  $|\rho| = 0.93 / 0.98$ )—while the Faithfulness and Localization metrics carried independent signal ( $|\rho| < 0.25$ ). Together, the structural pre-screen (Completeness, Non-Sensitivity, and the failed MPRT) and the greedy pruning of the two correlated pairs reduced the twelve-metric battery to an *identical* seven-metric set  $M^*$  on *both* models (faithfulness\_correlation, max\_sensitivity, pixel\_flipping, pointing\_game, random\_logit, relevance\_mass\_accuracy, sparseness). That two different architectures yield the same  $M^*$  shows the redundancy is a property of the metric battery rather than of the classifier, so an evaluator quoting “twelve metrics” is in effect quoting roughly seven—and can prune the rest with no loss of discriminative signal: the per-method ranking over the full computed battery and over  $M^*$  are identical on both models (Spearman  $\rho = 1.00$ ).

**6.1.0.0.2 Contribution 2 — Multi-criteria aggregation framework, and a test of reliability-discounted weighting.** The thesis implements a framework that integrates MetaQuantus meta-validation [14] with the Autoweighted aggregation of NormEnsembleXAI [11], and uses it to *test*—rather than merely assert—whether a MetaQuantus-discounted weighting operator (Equation (3.8)), which attenuates each metric’s weight by its measured reliability, changes the FAE ranking. Two findings result, and honesty requires stating both as they are. First, a positive but *conditional* one: how much the ranking changes with the weighting scheme is model-dependent. On ResNet-18 the uniform and Autoweighted method-rankings agree only moderately (Spearman  $\rho = 0.75$ ; Occlusion stays first under both, but Grad-CAM drops from second to fourth while Saliency and Integrated Gradients move up), whereas on SqueezeNet they agree almost perfectly ( $\rho = 0.96$ , Grad-CAM first under both). This is evidence that aggregation *can* materially reorder an FAE ranking—a warning against single-number leaderboards—but not that it always does; the most plausible mechanism is that the inverse-variance Autoweighted weights redistribute weight differently according to each model’s per-metric score-variance structure. Second, the proposed MQ-discount itself proves consequential in the same model-dependent way. Measuring it required extending the MetaQuantus Adversarial-Reactivity test to the two metrics that re-compute attributions internally (the degradation is injected into the explanation function itself, Section 4.3); with the extension, all seven  $M^*$  metrics carry a measured reliability. The completed discount reorders the ResNet-18 mid-field relative to plain Autoweighted ( $\rho = 0.79$ ; LRP rises from fourth into the top three—second at the point estimate—driven by the low measured reliability of random-logit and the high reliability of the max-sensitivity/sparseness/pixel-flipping trio) while leaving SqueezeNet nearly unchanged ( $\rho = 0.96$ ; mean shift  $|\Delta| \approx 0.07$ ). The contribution is the measured answer itself: a reliability-discounted aggregation *can* change the FAE verdict, whether it does is architecture-dependent, and no ground truth certifies the discounted ranking as the better one—the framework surfaces the sensitivity rather than resolving it.

**6.1.0.0.3 Contribution 3 — Empirical characterisation of metric ensembling.** The thesis uses the framework to characterise where aggregation of metric scores produces rankings that diverge from single-metric rankings and where it does not, and to compare individual against NormEnsembleXAI-ensembled attributions (Sections 4.4–4.5). The methodological outcome is an experimental protocol for isolating the effect of aggregation, and of attribution ensembling, on method ordering. *Verdict.* Both halves of this analysis return cautionary results. On the metric-aggregation side, a single-metric ranking (Faithfulness Correlation alone) is not a safe proxy for the multi-criteria index—the two correlate at only  $\rho = 0.29$  on ResNet-18 and 0.86 on SqueezeNet—confirming that the metric family chosen, and how it is weighted, can change which method wins. On the attribution-ensembling side, the result is *negative*. Over all seven  $M^*$  metrics and the full 600 images, with ensemble and individual scores normalised on a common (pooled) scale, the NormEnsembleXAI Mean ensemble beats none of the reported baselines: it is significantly worse than the *mean* individual method on ResNet-18 ( $-0.012$  vs.  $0.022$ ,  $p < 10^{-3}$ ), statistically indistinguishable from it on SqueezeNet ( $p = 0.74$ ), and decisively below the best *fixed* method, the best gradient-based member it actually contains, and the per-image oracle on both models ( $p < 10^{-3}$  throughout). A nominal SqueezeNet advantage reported under an earlier per-source normalization did not survive the common scale and is classified as a normalization artefact. Averaging attribution maps thus bought no effectiveness on this task; ensembling is not a substitute for identifying the per-model best method.

**6.1.0.0.4 Contribution 4 — Empirical validation on ISIC 2017.** The thesis validates the framework on expert-annotated dermoscopic data, using the segmentation masks to make the Localization category measurable, and applies Friedman, Nemenyi, and Wilcoxon testing to assess the significance of method and ensembling differences (Section 4.6). The methodological outcome is an end-to-end, statistically grounded instantiation of the framework on a real, mask-annotated medical task. *Verdict.* The differences the framework reports are statistically real, not sampling noise. The Friedman omnibus test over the seven methods and 600 image-blocks rejects equal mean ranks for every scheme and model at  $p < 10^{-3}$  (ResNet-18  $\chi^2 = 760$  Autoweighted / 469 MQ-discount; SqueezeNet  $\chi^2 = 1430$  / 1218; Kendall’s  $W$  between 0.13 and 0.44), with a Nemenyi critical difference of  $CD = 0.368$ . Five of the six paired Wilcoxon ensemble comparisons are significant at  $p < 10^{-3}$ ; the sixth—ensemble vs. the mean individual method on SqueezeNet—is reported as a null ( $p = 0.74$ ), which is itself informative. The statistics therefore establish that the negative ensemble result and the weighting-driven ranking divergence are genuine effects of the framework’s configuration rather than artefacts of the particular draw of images—which is precisely what makes the cautionary findings above defensible.

Taken together, the four contributions answer the thesis objective not with a single “best” FAE method—which would be dataset- and configuration-specific and of limited durability—but with a principled, reproducible apparatus for reaching such a verdict, and with an honest empirical characterisation of where that apparatus changes a conclusion (substantial metric redundancy; a model-dependent ranking divergence under different weightings, including the reliability discount itself) and where it does not

(ensembling, which beats none of the individual baselines once scores share a common scale). The yield is deliberately cautionary: its main practical lesson is to distrust single-number FAE leaderboards and to test, rather than assume, that any aggregation refinement earns its place.

## 6.2. Future Work

The framework was deliberately bounded in scope (Chapter 5), and each boundary suggests a concrete extension. The directions below are ordered roughly from those that consolidate the present results to those that broaden the framework’s reach.

**6.2.0.0.1 Scaling the statistical analysis.** The redundancy and reliability estimates rest on finite samples from a single challenge distribution—the redundancy analysis on the full 600-image test set, the meta-evaluation on a 64-image sample (12 for the extended-AR completion). The immediate next steps are to enlarge the meta-evaluation sample and, more importantly, the number of degradation levels (the 5-level AR estimator has 0.1-granularity regardless of image count); to repeat both analyses on repeated subsamples, reporting confidence intervals on the correlation and reliability estimates and a stability analysis of the membership of  $M^*$  and of the metric weights; and to adopt a held-out-split protocol that separates metric selection and weighting from significance testing (Chapter 5). This would move the metric structure from a property of one draw of images to a property of the domain. Validating the extended Adversarial-Reactivity protocol belongs to the same consolidation: the AR scores of *max\_sensitivity* and *random\_logit* are measured under this thesis’s degraded-explainer extension rather than the original MetaQuantus array-degradation test, and comparing the extension against a reference implementation (or an alternative operationalisation of “explanation degradation” for such metrics) would firm up the reliability estimates that drive the ResNet-18 reordering. A related refinement concerns the pruning policy: on this dataset the greedy cross-category rule of Decision D4 and its strictly within-category alternative happen to coincide (both pruned pairs lie within one category), so verifying whether that equivalence survives on datasets with a different cross-category correlation structure remains open.

**6.2.0.0.2 Automated meta-validation.** Meta-validation is currently a configured stage applied once to a fixed metric set. A natural extension is to fold it into a fully automated loop that, given a new dataset or model, re-derives the reliable, non-redundant metric subset and the MQ-discounted weights without manual intervention. This would turn the pipeline from a study-specific configuration into a reusable “evaluate-the-evaluators” service, and would let the framework adapt its metric set to each new domain rather than assuming the ISIC 2017 structure transfers.

**6.2.0.0.3 More datasets and architectures.** All evidence here comes from ISIC 2017 dermoscopy and from two convolutional classifiers. Extending the study to a second mask-annotated medical dataset would test whether the redundancy and reliability structure transfers across domains, while adding a

transformer-based classifier would test whether gradient-based attributions and the layer-wise MPRT behave differently outside the convolutional regime. Both extensions directly probe the external-validity limits identified in Chapter 5.

**6.2.0.0.4 Additional FAE methods, including the deferred perturbation methods.** The evaluated panel was restricted to seven methods for reasons of computational cost and metric compatibility; LIME and KernelSHAP were deferred rather than dismissed. Re-introducing the perturbation-based methods—and taming the cost of the Robustness metrics’ repeated re-explanation of every perturbed input, perhaps via caching or cheaper surrogate refits—would test whether the framework’s conclusions hold across the gradient/perturbation divide and would let the redundancy analysis speak to whether perturbation-based faithfulness metrics are redundant with their gradient-based counterparts.

**6.2.0.0.5 Aggregation operators and weighting.** The thesis recommends the weighted arithmetic mean but reports the minimum and geometric-mean operators as alternatives encoding worst-dimension and catastrophic-failure-averse stances respectively. A focused study could compare these operators under the Autoweighted weighting (and the MQ-discount, which materially reorders the ResNet-18 ranking here), particularly for safety-critical interpretations where the minimum operator’s “acceptable-on-every-axis” semantics are most defensible, and could explore learned or preference-elicited weightings as a successor to the data-adaptive Autoweighted scheme. On the attribution-ensembling side, replacing the fixed Mean combinator with an *optimized* convex combination of attributions [27] is the natural next step, given that the plain ensemble never beat the best individual method in this study.

**6.2.0.0.6 Regulatory alignment and auditability.** Finally, the reproducible, meta-validated pipeline is well positioned to serve as a transparency instrument under emerging regulation. Following Seth and Sankarapu [36], a promising direction is to connect the framework explicitly to the transparency and documentation requirements of the EU AI Act, formalising the pipeline’s configuration record as an auditable evaluation artefact and assessing what additional evidence—human-grounded validation in particular—a regulator would require before a functionally grounded effectiveness index could support a compliance claim. Systematic reviews of the law-to-XAI translation gap confirm that the Act mandates transparency without prescribing operational metrics [45], and first frameworks that score XAI methods directly against the Act’s explainability requirements are emerging [46]—precisely the gap that a meta-validated effectiveness index is positioned to fill. This would extend the contribution from a research methodology toward a candidate evaluation standard.



## 7. List of Symbols and Abbreviations

This chapter collects the abbreviations and mathematical notation used throughout the thesis. The abbreviations are listed first, followed by the mathematical symbols introduced in the formalisation of the evaluation framework (Chapters 2 and 3).

### Abbreviations

**AI** – Artificial Intelligence

**XAI** – Explainable Artificial Intelligence

**DL** – Deep Learning

**CNN** – Convolutional Neural Network

**FAE** – Feature Attribution Explanation

**IG** – Integrated Gradients

**Grad-CAM** – Gradient-weighted Class Activation Mapping

**LRP** – Layer-wise Relevance Propagation

**LIME** – Local Interpretable Model-agnostic Explanations

**SHAP** – SHapley Additive exPlanations

**AUC** – Area Under the Curve

**ReLU** – Rectified Linear Unit

**SGD** – Stochastic Gradient Descent

**PG** – Pointing Game (Localization metric)

**RMA** – Relevance Mass Accuracy (Localization metric)

**FC** – Faithfulness Correlation (Faithfulness metric)

- MPRT** – Model Parameter Randomisation Test (Randomization metric)
- NR** – Noise Resilience (MetaQuantus meta-evaluation test)
- AR** – Adversarial Reactivity (MetaQuantus meta-evaluation test)
- SMS** – Second Moment Scaling (normalization transformation)
- ES** – Ensemble Score (per-metric ensemble agreement)
- MQ** – MetaQuantus (meta-evaluation of metrics)
- CD** – Critical Difference (Nemenyi post-hoc significance)
- ISIC** – International Skin Imaging Collaboration (dataset)
- EU** – European Union
- GDPR** – General Data Protection Regulation
- CSV** – Comma-Separated Values
- YAML** – YAML Ain't Markup Language (configuration format)

## Mathematical Symbols

- $f$  – Predictive model (classifier) whose decisions are explained
- $x$  – Input image (instance) presented to the model
- $x'$  – Baseline input used by attribution methods (e.g. a black image)
- $c$  – Target class index for which an explanation is produced
- $\hat{y}$  – Model's top-1 prediction,  $\hat{y} = \operatorname{argmax}_c f(x)_c$ , used as attribution target
- $\Phi$  – Feature attribution explanation method (estimator)
- $\Phi(f, x)$  – Attribution map produced by method  $\Phi$  for model  $f$  and input  $x$
- $E$  – Explanation (attribution) map,  $E = \Phi(f, x)$ ,  $E \in \mathbb{R}^d$
- $E_i, R_i$  – Relevance score assigned to input feature  $x_i$
- $d$  – Dimensionality of the input (number of features / pixels)
- $C, H, W$  – Channel, height, and width dimensions of an attribution map  $\Phi(f, x) \in \mathbb{R}^{C \times H \times W}$
- $\mathcal{M}$  – Binary segmentation (lesion) mask used by Localization metrics

- 
- $\mathcal{D}$  – Test set over which metrics and correlations are computed
  - $M_k$  – The  $k$ -th evaluation metric (raw score  $M_k(\Phi)$ )
  - $K$  – Number of evaluation metrics under consideration
  - $\{M_1, \dots, M_K\}$  – Full set of available evaluation metrics
  - $M^*$  – Reduced, non-redundant and meta-validated subset of metrics
  - $E(\Phi)$  – Aggregated effectiveness index of method  $\Phi$
  - $s_k$  – Direction sign of metric  $k$ ,  $s_k \in \{+1, -1\}$  (+1 = higher-is-better)
  - $\hat{M}_k$  – Direction-rectified score,  $\hat{M}_k = s_k \cdot M_k$
  - $\psi_k$  – Scale-rectifying normalization transformation for metric  $k$
  - $\tilde{M}_k$  – Normalized (rectified and scaled) score of metric  $k$
  - $\mathcal{N}$  – Normalization function applied to attributions before ensembling
  - $\mathcal{A}$  – Aggregation (combinator) operator over normalized scores
  - $w_k$  – Weight assigned to metric  $k$  in the aggregation,  $w_k \geq 0$ ,  $\sum_k w_k = 1$
  - $\mathbf{w}$  – Weight vector  $\mathbf{w} = (w_1, \dots, w_K)$
  - $cv_k$  – Coefficient of variation of metric  $k$ 's raw scores across the test set
  - $a_k$  – Autoweighted agreement weight of metric  $k$ ,  $a_k = 1/(cv_k + \varepsilon)$
  - $w_k^{\text{AW}}$  – Normalized Autoweighted coefficient of metric  $k$
  - $\text{NR}_k$  – Noise Resilience score of metric  $k$  from meta-evaluation
  - $\text{AR}_k$  – Adversarial Reactivity score of metric  $k$  from meta-evaluation
  - $\lambda_k$  – Combined-reliability coefficient of metric  $k$ ,  $\lambda_k \in [0, 1]$  (meta-evaluation discount)
  - $\rho$  – Correlation measure (Spearman/Pearson) used in MPRT and Random Logit
  - $\rho_{ij}$  – Empirical correlation between metrics  $M_i$  and  $M_j$  over  $\mathcal{D}$
  - $r(\cdot, \cdot)$  – Pearson correlation coefficient
  - $\mu_k, \sigma_k$  – Group mean and standard deviation of metric  $k$  ( $z$ -score normalization)
  - $\varepsilon$  – Small positive constant (geometric-mean stabiliser / perturbation bound)
-

$\delta_j$  – Random input perturbation (uniform noise of bounded magnitude) in robustness metrics

$\alpha_k^c$  – Grad-CAM neuron-importance weight for feature map  $k$  and class  $c$

$\mathbb{E}[\cdot]$  – Expectation operator

$\mathbb{R}$  – Set of real numbers

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